

**The Neurocognition of Longitudinal L3 Morphosyntactic Transfer: A Bayesian Single-Trial  
ERP Study of Three Grammatical Properties in Two Artificial Languages**

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### Abstract

Theories of third-language (L3) morphosyntactic transfer disagree over whether prior-language influence is wholesale or property-by-property, and are silent on how it consolidates. We tracked electrophysiological and behavioural signatures of three properties (subject–verb gender agreement, differential object marking, verb–object number agreement) as Norwegian–English bilinguals acquired Mini-English or Mini-Norwegian over four sessions, from the first post-training session to post-consolidation retention. Single-trial event-related potential (ERP) amplitudes and judgement accuracy were analysed in Bayesian multilevel models with informative priors. Gender agreement, whose Mini-Norwegian realisation aligns with the learners’ first language, produced a grammaticality effect that grew with training. In a refit that freed the retention session from that trend, the effect at retention did not depart credibly from the trend on average, but the two languages diverged credibly, the Mini-English estimate falling below the Mini-Norwegian one. Only the group difference is supported. Behaviourally, the same property favoured Mini-Norwegian learners, the design’s only first-language-congruent advantage. Differential object marking yielded a transient early effect, a late language-conditioned modulation and a behavioural decline at retention. Verb–object number agreement, novel to both prior languages, strengthened across the retention interval, and within groups only for Mini-Norwegian learners. In whole-scalp decoding, only gender agreement separated the conditions on a baseline at chance. Descriptively, no property’s pattern generalised to another, consistent with property-specific coding. Both novel properties showed source-language modulations of opposite sign, unexplained. The growth of the gender-agreement advantage across sessions favours a property-by-property account over a wholesale one, though one dissociating property leaves the conclusion provisional.

*Keywords:* L3 acquisition, morphosyntactic transfer, ERP, artificial language, Bayesian multilevel models, retention

## **The Neurocognition of Longitudinal L3 Morphosyntactic Transfer: A Bayesian Single-Trial ERP Study of Three Grammatical Properties in Two Artificial Languages**

When multilingual speakers add a third or later language (L3/Ln), the grammars they already command shape how its morphosyntax is represented and processed. How that prior knowledge is recruited is the central question dividing contemporary models of L3 morphosyntactic transfer. Wholesale-transfer accounts hold that a single previously acquired grammar is selected as the basis for the initial L3 grammar, whether on typological grounds (the Typological Primacy Model: [Rothman, 2011, 2015](#)) or because the second language enjoys a privileged cognitive status (the L2 Status Factor: [Bardel & Falk, 2007](#); [Falk & Bardel, 2011](#)). Property-by-property accounts predict instead that transfer is selective and incremental, with individual structures transferred (and, where helpful, “enhanced”) one at a time as the learner detects structural similarity. The Linguistic Proximity Model ([Westergaard et al., 2017, 2023](#); [Westergaard, 2021](#)), the Scalpel Model ([Slabakova, 2017](#)) and the Cumulative-Enhancement Model ([Flynn et al., 2004](#)) all take that form. The wholesale position has recent defenders who raise conceptual and empirical objections to selective transfer ([Schwartz & Sprouse, 2021](#)), and initial-stage evidence from early Catalan–Spanish bilinguals, whose performance on four properties was compatible with Catalan as the single source of their initial English grammar, has been read as support for it ([González Alonso et al., 2021](#)). Any test that discriminates between them requires (a) multiple, independently manipulable grammatical properties, (b) a source language whose relationship to each property is controlled, and (c) a temporal window long enough to watch transfer emerge and to test whether what transferred is retained.

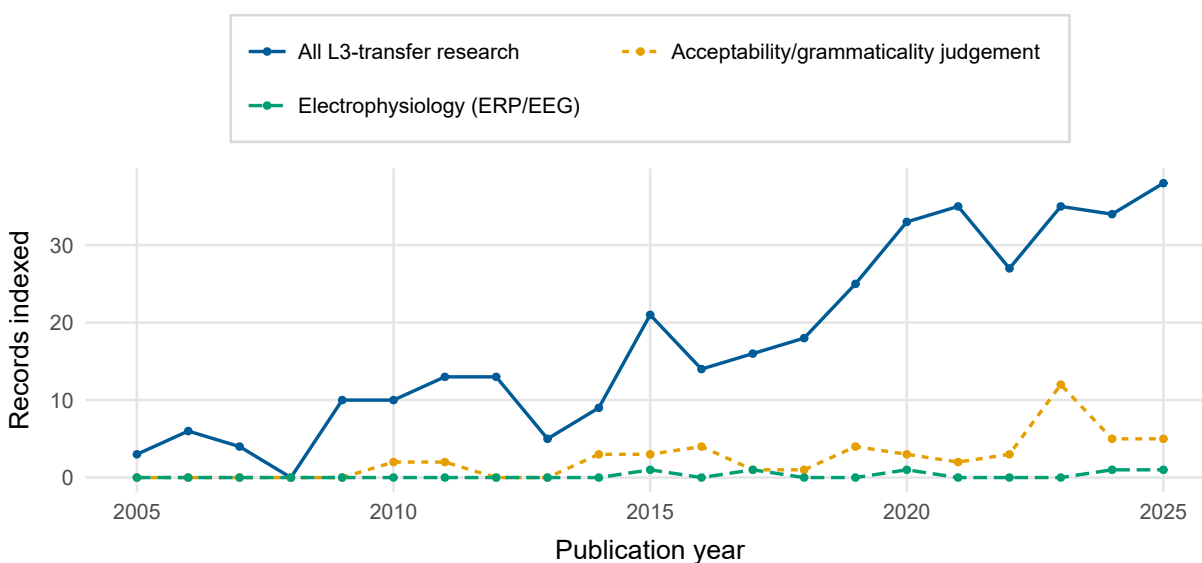
### **Electrophysiology of morphosyntactic processing**

Event-related potentials (ERPs) provide a millisecond-resolution index of how the brain responds to grammatical violations and are therefore well suited to revealing the representations that underlie transfer before they surface in overt behaviour. Most evidence on L3 morphosyntactic transfer has come from offline, untimed acceptability judgements ([Puig-Mayenco et al., 2020](#); [Skafba et al., 2026](#)). These index the outcome of learning. They leave untouched the

moment-to-moment processing that ERPs expose (Angelovska & Roehm, 2023; Rothman et al., 2015). A single-trial ERP design therefore tests directly whether source-language structure modulates online comprehension. Online evidence has begun to appear. Eye-tracking in early learners of Norwegian shows facilitative and non-facilitative influence from each prior language on different properties (Skafba et al., 2026). The imbalance is visible in the literature itself (Figure 1). In all, 369 records on L3 transfer were indexed between 2005 and 2025. Of those, 50 used acceptability or grammaticality judgements and 5 used electrophysiology, a ratio of roughly 10 to one. The field has grown steadily while its electrophysiological base has not.

**Figure 1**

*Methodological Composition of the L3 Morphosyntactic-Transfer Literature Indexed in Scopus, by Publication Year*



*Note.* The upper line is all L3-transfer research. The lower two are the subsets whose title, abstract or keywords mention acceptability or grammaticality judgements, and electrophysiology (ERP/EEG). The counts index relative composition and are not an exhaustive census: Scopus under-covers books and chapters, keyword matching errs in both directions, and records indexed late may still be added for the most recent years. The retrieval queries are recorded in the project repository.

Three components are relevant here. The left-anterior negativity (LAN) is often linked to automatic morphosyntactic agreement checking. The N400, a negativity sensitive to lexical–semantic and thematic integration (Frisch & Schlesewsky, 2001; Nieuwland et al., 2013),

also marks the early stages of morphosyntactic learning, before a late positivity emerges with rising proficiency ([Osterhout et al., 2006](#)). That later positivity is the P600, first reported for words inconsistent with the preferred syntactic analysis ([Osterhout & Holcomb, 1992](#)). Over centro-posterior sites, it has since been associated with syntactic reanalysis and repair, and with the late integration of syntactic and semantic information ([Friederici, 2002](#); [Friederici et al., 2002](#)). These map onto the three latency windows analysed here (200–500, 300–600 and 400–900 ms). Training studies show that the brain’s response to second-language violations becomes more native-like as proficiency increases ([Morgan-Short, Steinhauer, et al., 2012](#)), and again across a consolidation interval containing no further exposure ([Morgan-Short, Finger, et al., 2012](#)), with negativities and P600s emerging or shifting over time. Native speakers ([Tanner & Van Hell, 2014](#)) and learners ([Tanner et al., 2013](#)) also differ systematically from one another in whether they recruit a more negativity-dominated or a more P600-dominated response. Artificial-language paradigms, in which the input is fully controlled, have been particularly valuable for isolating these dynamics from the confounds of natural-language exposure ([Friederici et al., 2002](#); [González Alonso et al., 2020](#)). Bilinguals learning such a language show a P600 to syntactic violations at a lower proficiency than monolinguals do ([Grey et al., 2018](#)).

The closest antecedent to the present design is the artificial-language L3 ERP work of González Alonso et al. ([2020](#)). Its findings shape both our predictions and our reading of the results, so we set them out in full. Fifty first-language-Spanish, second-language-English bilinguals were given implicit training on one of two miniature languages, built on English and Spanish lexicons, and their ERPs to gender-agreement violations were recorded at first exposure. They named their two languages after those lexicons, and we refer to their groups by lexicon throughout, so that neither is confused with our own Mini-English and Mini-Norwegian conditions. The two groups responded differently. In their English-lexicon language, violations elicited an early (200–500 ms) fronto-lateral negativity, confined to electrodes in the right-anterior region and followed by nothing in later windows. In their Spanish-lexicon language, they elicited a broadly distributed positivity at 300–600 ms. Neither group produced a P600 or any other canonical index of grammatical

processing. The authors read the between-group difference as differential allocation of attention, driven by each miniature language's lexical similarity to a prior grammar, and therefore as a precursor of source selection, not as grammatical processing itself. The present study extends that logic in three ways. It goes beyond first exposure, tracking each property across up to four sessions, and it manipulates three properties whose relation to the prior grammars differs within the same learners. It also closes with an unrehearsed retention test after months without exposure, so the durability of what transferred can be measured. Whether the initial-exposure pattern of González Alonso et al. (2020) reappears under these conditions is itself an empirical question, taken up in the Discussion.

Two behavioural studies share more of the present materials. Each trained speakers on a miniature language with a Norwegian lexicon and a case system, a property Norwegian lacks. Prior knowledge of a case language eased acquisition even though the lexicon pointed to Norwegian, but only where that language and the miniature language marked case in the same overt form, on the noun (Castle et al., 2026; Mitrofanova et al., 2023). Greek, which marks case on the determiner, gave no such advantage in the first study, and the second found the advantage subject to word-order bias and proficiency. The overt realisation of a property conditioned its transfer, and the present gender manipulation varies exactly that realisation between the two miniature languages. Recent syntheses of the transfer literature draw attention to the methodological choices that constrain what its findings can show (Alonso et al., 2026; Puig-Mayenco et al., 2020; Rothman et al., 2019), and longitudinal and electrophysiological evidence remains among its scarcest. An ERP successor to González Alonso et al. (2020) trained heritage and second-language bilinguals on a Latin mini-grammar and found violation responses that no current model had predicted, an outcome its authors took as a question about the paradigm itself (Pereira Soares et al., 2022). The recurring objection to miniature languages is that a grammar acquired in hours may not engage the mechanisms of natural morphosyntax. This literature answers it with the finding that artificial-language learners, once sufficiently proficient, show qualitatively native-like violation responses (Friederici et al., 2002; Morgan-Short, Steinhauer, et al., 2012). What the design buys in

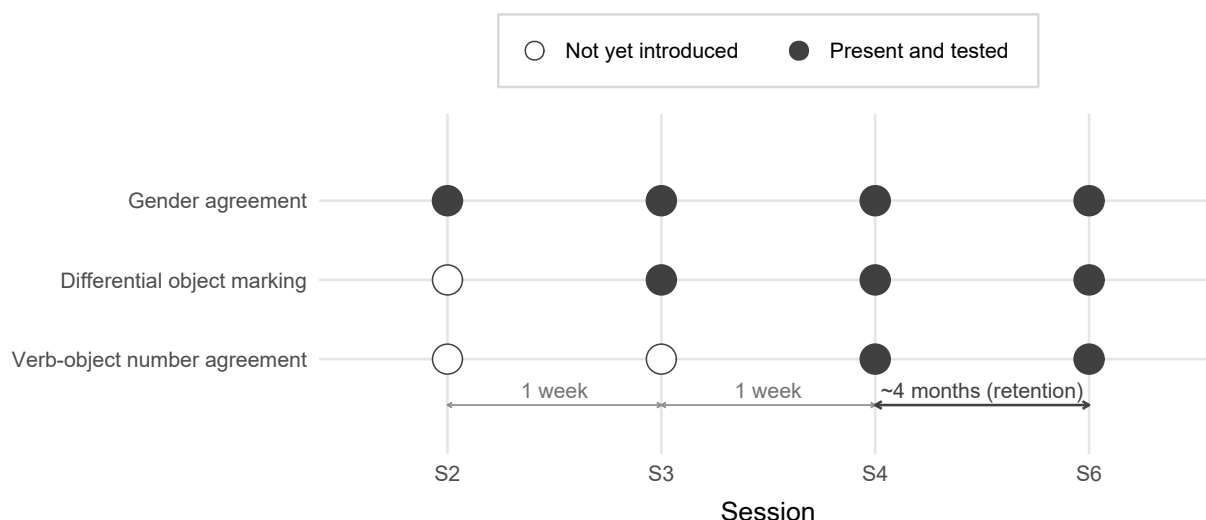
exchange is exact control over every learner's input history, which no natural-language design can offer.

### **The present study**

We exploit a controlled artificial-language design to test predictions that separate wholesale from property-by-property transfer, and to characterise the time course of transfer from the first post-training session through post-training retention. Norwegian–English bilinguals (first language Norwegian, second language English) acquired one of two miniature languages, Mini-English or Mini-Norwegian. Within each language, we manipulated three morphosyntactic properties that vary in their relationship to the learners' existing grammars: subject–verb gender agreement, differential object marking and verb–object number agreement. The two languages share their morphemes but differ in how they realise each property. Mini-Norwegian expresses gender agreement through a suffixed determiner, as Norwegian does, whereas Mini-English uses an English-like free determiner. The same property therefore aligns with the first language for one group only (see Method).

Single-trial ERPs and grammaticality judgements were collected across four laboratory sessions (Sessions 2, 3, 4 and 6) that trace the learning trajectory from the first post-training session to a post-consolidation retention test. The properties enter the design at different points (gender agreement from Session 2, differential object marking from Session 3 and verb–object number agreement from Session 4), yielding a cumulative, property-by-property exposure schedule (Figure 2).

Three quantities operationalise our hypotheses. The grammaticality effect (the amplitude or accuracy difference between ungrammatical and grammatical trials) indexes whether a property is processed grammatically at all. The grammaticality  $\times$  session interaction indexes *learning*: the sharpening of that effect with exposure. Its modulation by language, the grammaticality  $\times$  session  $\times$  language term, indexes *source-language transfer*, whether learning is faster or stronger for the language whose prior structures align with the target property. A derived Session 4  $\rightarrow$  Session 6 contrast then indexes how the grammaticality effect changes across the consolidation interval. The language advantage evaluated at a property's first session, derived from the same draws, indexes

**Figure 2***Exposure and Measurement Schedule*

*Note.* Each grammatical property enters at a different session and recurs thereafter, and the four ERP/behavioural sessions are 2, 3, 4 and 6. Gender agreement therefore has four sessions of coverage, differential object marking three, and verb–object number agreement two. The session axis is ordinal, so equal spacing does not imply equal elapsed time. Sessions 2 to 4 are one week apart, whereas Session 6, the unrehearsed retention test, follows Session 4 after about four months without exposure. The intervals are therefore annotated numerically.

whether that advantage was in place when the property was first probed.

The transfer accounts make no prediction about the retention interval, since they concern the initial state and the course of acquisition. The expectation for that interval comes from the training literature cited above, in which a grammaticality response that has emerged during training holds or grows across months without exposure (Morgan-Short, Finger, et al., 2012). If transfer is property-specific, that profile may differ between the property with a first-language analogue and the two without. The retention contrasts therefore characterise the trajectory. They do not test between the accounts.

The two families of account are separated here by which prior grammar each engages and by when its influence should appear. Gender agreement is the only property with a first-language analogue, and so the only one on which the two miniature languages stand in different relations to a prior grammar. Differential object marking and verb–object number agreement are novel to

Norwegian and English alike and are realised identically in the two miniature languages. Property-level congruence therefore predicts no group difference on them under either family, and they serve as controls. A wholesale account, on which one prior grammar supplies the initial state for the L3 grammar as a whole, must also say which grammar is selected. The Typological Primacy Model selects the source from a rapid typological assessment led by lexical similarity, and the design speaks to that cue directly. Each miniature language draws its content words from one of the prior languages, so lexically led selection points each group to the language its miniature language is built on, English for Mini-English and Norwegian for Mini-Norwegian. On that selection, a wholesale account predicts a Mini-Norwegian advantage on gender agreement, present from that property's first session, because the projected Norwegian grammar supplies both the gender system and its suffixal format. It predicts no group difference on the control properties, which neither projected grammar encodes. A property-by-property account predicts the same confinement of the advantage to gender agreement but a different onset, with the advantage emerging only as the learner detects the structural match. Under the source selection this design supports, then, the two families agree on where the gender-agreement advantage should appear and differ only on when. The diagnostic quantity is therefore the advantage at the property's first session, which wholesale projection predicts to be present and a property-by-property account predicts to be absent or small. Growth of the advantage across sessions is predicted by the property-by-property account and permitted by wholesale projection, which constrains the initial state and says nothing about subsequent learning ([Alonso & Rothman, 2017](#)), so growth alone favours the former without excluding the latter.

The L2 Status Factor makes a further prediction of its own. If the second language is the default source, both groups project English, which encodes none of the three properties, so property-level congruence predicts no group difference anywhere. If projecting a source whose morphological format matches the input eases processing of that input at all, the beneficiary under an English default is the Mini-English group, whose miniature language is built on English-like free morphology. In no derivation does an L2 default predict a Mini-Norwegian advantage. This design

discriminates the two families less sharply than a design contrasting two candidate source grammars would, and we return to the point in the Limitations.

Alongside the region-averaged models, we ask whether grammaticality is decodable from the whole-scalp voltage pattern at each moment. A multivariate decoder can detect a distributed violation response too weak to move any single region's mean. Its cross-language generalisation asks whether the pattern that separates the conditions in one mini-language also separates them in the other. Its cross-property generalisation tests a further prediction. If the grammaticality response were largely a domain-general salience or integration signal (Aurnhammer et al., 2023; Sassenhagen & Fiebach, 2019), a decoder trained on one property should transfer to the others. If each property recruits its own code, as the fractionation of the late positivity (Kaan & Swaab, 2003) and the separable developmental courses of agreement features (Alemán Bañón et al., 2018) suggest, it should not. No null distribution was computed for either generalisation analysis, so both are reported as descriptive probes.

## Method

### Participants

In total, 65 Norwegian–English bilinguals (first language Norwegian, second language English) took part, each assigned to learn one of the two miniature languages. Assignment followed a fixed rule on the participant identifier, giving 33 Mini-English and 32 Mini-Norwegian learners, so language is a between-participants factor. Of the 64 participants with a Language History Questionnaire record (LHQ3: Li et al., 2020), the mean age was 25.7 years ( $SD = 7.4$ , range 16–54). The youngest participant was under 18 at the time of the questionnaire. All others were adults. Of the same record, 40 identified as female, 21 as male and 1 as non-binary (2 did not disclose). Among the 63 who reported handedness, 60 were right-handed and 3 left-handed. English had been acquired early (age of first exposure  $M = 6.8$  years,  $SD = 4.0$ ) and used for many years ( $M = 18.2$ ,  $SD = 7.5$ ), with high self-rated proficiency ( $M = 0.82$  of 1,  $SD = 0.16$ ). 48 of the 64 also reported knowledge of at least one further language, most often German or Spanish. Several of the languages named mark grammatical gender or case, and the design did not screen them out (see Limitations).

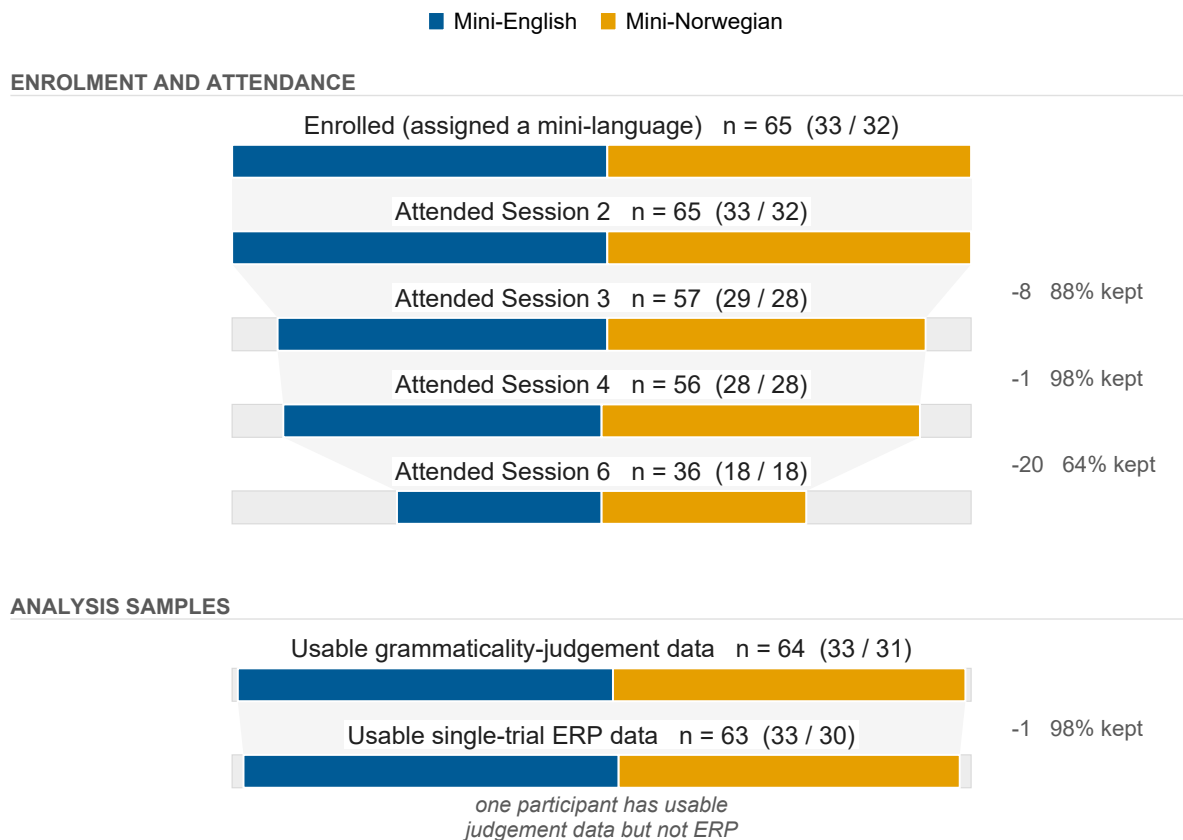
Attrition increased across the training: of the 65 who began, 57 remained at Session 3, 56 at Session 4, and 36 completed the Session 6 retention test. The ERP analyses draw on the 63 participants with usable single-trial EEG (33 Mini-English, 30 Mini-Norwegian). The fitted ERP models also require the L1-dominance covariate described under *Data analysis*, and the trials of a participant without that score are dropped by listwise deletion, so the gender-agreement models were fitted to 62 participants, the differential-object-marking models to 56 and the verb–object number-agreement models to 55. The grammaticality-judgement analyses draw on 64 participants (Figure 3). The project was registered with the Norwegian Agency for Shared Services in Education and Research (Sikt; project 2559611). All procedures complied with the relevant laws and institutional guidelines, the privacy rights of the participants were observed, and every participant gave written informed consent.

### Design and materials

The longitudinal artificial-language paradigm implemented here is that of the wider project reported by González Alonso et al. (2025). Within each language, the design crosses grammaticality (contrast coded grammatical = +0.5, ungrammatical = −0.5) with session (the cumulative exposure schedule above) within participants, and language between participants (Mini-Norwegian = +0.5, Mini-English = −0.5), so a positive language-bearing coefficient indicates a larger value for the Mini-Norwegian group. The three properties were analysed separately. Each was probed with 48 grammatical and 48 ungrammatical sentences per session, built from a closed lexicon of twelve nouns, twelve adjectives, twelve transitive verbs and twelve proper names per language. A third, ancillary violation type was recorded but is not analysed here. Each sentence frame was presented in both a grammatical and an ungrammatical version. The two miniature languages draw on separate content lexicons, Mini-English building its nouns, adjectives, verbs and names on English stems and Mini-Norwegian on Norwegian stems. The critical word carried the EEG trigger that defined the single-trial time-lock.

The two miniature languages share the same invented grammatical morphemes (the determiners *jel/zel/jer/zer*, the object marker *fi*, the verbal suffix *-evo* and the adjectival agreement

**Figure 3**  
*Participant Flow*



*Note.* Each stage spans the full enrolled cohort: the solid block is the participants still present, split by mini-language, and the faint block completing the bar is those no longer present, so cumulative attrition is visible as the unfilled area. Tapering bands connect consecutive stages, and each transition is annotated with the number lost and the percentage retained. The upper tier tracks enrolment and attendance at the four sessions this paper analyses. Sessions 1 and 5 are online cognitive sessions belonging to the companion paper and are omitted. The lower tier holds the analysis samples, which draw on the whole cohort pooled across sessions and not on Session 6, so bands are drawn only within a tier, where the subset relation holds. Two features of the cohort prevent a strictly nested reading: one participant missed Session 5 and returned for Session 6, and one contributes usable grammaticality judgements but no usable ERP.

suffixes). They differ in one structural dimension, determiner position, and that difference is the pivot of the transfer contrast. Subject–verb gender agreement is built on a two-way common/neuter distinction, as in real Norwegian. In Mini-English, the determiner is a free prenominal word that marks gender and number (*je/ze* for common/neuter singular, *jer/zer* for the plural), as English articles are free-standing. In Mini-Norwegian, the same gender–number determiner is a bound suffix on the noun (e.g. *gateje/gatejer*), as the Norwegian definite article is suffixal. In both languages, the predicate adjective carries a matching agreement suffix, on which the gender-agreement violation is realised. Differential object marking is a free particle *fi* placed before the object noun phrase, identical in the two languages, with the violation omitting it. Verb–object number agreement is a verbal suffix *-evo* that marks the verb to agree with a plural object, again identical across languages, with the violation a number mismatch between verb and object.

These three properties map onto the learners’ existing grammars differently, which is what the language factor exploits. Grammatical gender expressed through a suffixed determiner is a property of the learners’ first language, Norwegian, but not of their second language, English. The Mini-Norwegian realisation therefore aligns closely with the first-language grammar, whereas the Mini-English realisation expresses a feature that English does not encode (gender) through an English-like free determiner. Differential object marking and verb–object number agreement, by contrast, are absent from both Norwegian and English, so neither is supported by a prior grammar in either group.

## Procedure

Sentences were presented one word at a time in the centre of the screen (rapid serial visual presentation, RSVP) using OpenSesame 3.3.14. Word duration was a deterministic function of word length. A non-critical word was shown for  $250 + 35 \times (\text{characters} - 3)$  ms, floored at 250 ms, giving durations of 250–600 ms. The critical word was shown for 40 ms less than a non-critical word of the same length (210–560 ms). Non-critical words followed one another at a stimulus-onset asynchrony of approximately 505 ms. At the critical word, the script held the asynchrony fixed at approximately 750 ms, lengthening the following blank interval to compensate for the varying word

duration. The timing of the post-critical region therefore did not depend on the length of the critical word. The critical word occupied the third or fourth position and was followed by three to six further words. Presentation was entirely experimenter-paced. Nothing in the sentence stream depended on the participant, so a condition difference in reading speed cannot shift the epoch. The grammaticality judgement was requested only after the final word of the sentence, never during it, and was made with a two-button response (left = grammatical, right = violation). No motor response therefore fell inside the analysed epoch.

This timing has one consequence for the late window. The raw presentation logs record all 64,907 critical-word trials, a census taken before any EEG-side exclusion, so its count exceeds the modelled trial numbers. In that census, exactly one subsequent word onset falls within 900 ms of the critical word (the per-trial maximum is 1), at 743–864 ms (median 747 ms). The final roughly 150 ms of the 400–900 ms measurement window therefore contains the beginning of the response to the following word. This is unavoidable in connected-sentence RSVP at this rate, and it is constant across conditions because the post-critical asynchrony is held fixed, so it cannot manufacture a grammaticality effect. Late-window amplitudes should nonetheless be read as containing that overlap. They are not a pure response to the critical word.

### **EEG acquisition and single-trial ERP quantification**

Continuous EEG was recorded with a 32-channel Brain Products LiveAmp system, using active electrodes in an extended 10–20 layout with the CACS-32 cap montage recorded in the data headers. The sampling rate was 500 Hz and the online passband DC–131 Hz, with no online notch filter. The 131 Hz anti-aliasing low-pass is a causal acquisition filter, so it imposes no phase distortion of the kind that motivates the offline filter choices set out below.

Three channel counts appear in what follows, and they refer to different things, so we set them out once. Thirty-two electrodes were recorded. Two of them, TP9 and TP10, become the offline linked-mastoid reference and are therefore not analysed as signals. FCz served as the online reference and is recovered as a channel when the data are re-referenced. The remaining 30 electrodes are the ones assigned to the nine analysis regions, so they are what the region-averaged

amplitude models use and what the scalp maps interpolate across. The decoder ignores the region map and takes every EEG channel available in all contributing sessions, which is those 30 plus the recovered FCz, giving 31 features. The LiveAmp's three accelerometer traces (`x_dir`, `y_dir`, `z_dir`) are movement signals, not brain signals, and are excluded throughout. The Session 2 exports additionally retain TP9, TP10 and the accelerometer channels, whereas the Session 3, 4 and 6 exports do not, which is why the decoder uses only the channels common to every session. Offline, in BrainVision Analyzer, the data were re-referenced to the mastoids and then band-pass filtered with zero-phase-shift (forward and reverse) Butterworth IIR filters, high-pass 0.1 Hz and low-pass 30 Hz, each specified in Analyzer as fourth order (24 dB per octave). Because the filters are applied in both directions, the effective roll-off is approximately double that figure. No notch filter was applied offline. Filtering preceded ocular correction, so the independent component analysis was trained on the same 0.1 Hz high-passed data that the analyses themselves use. Owing to a processing oversight, the Session 3 recordings of four participants (lab IDs 7, 8, 9 and 16) were high-pass filtered at 1 Hz rather than 0.1 Hz. These four participant-sessions are excluded from every fitted ERP model and from the decoding. They are retained in two places, both flagged where they occur. The accuracy models draw on the behavioural record, which the filter setting does not touch. The descriptive grand-average waveforms, difference waves and topographies shown in the Results are extracted on the full sample, and their captions say so. The exclusion rests on the filter setting itself, a documented generator of artefacts in this kind of paradigm ([Tanner et al., 2015](#); [Widmann et al., 2015](#)), and on nothing observed in these participants' data, so the choice is not results-contingent. The argument is set out in [Appendix A](#). The excluded datasets hold 6.2% for gender agreement and 6.2% for differential object marking of the retained Session 3 trials. The pipeline can be re-run with them retained.<sup>1</sup> Ocular artefacts were then corrected by independent component analysis, and the record was segmented into epochs from –100 to 1098 ms relative to

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<sup>1</sup> The repository documents a configuration switch that refits the ERP grid with the four datasets retained, so the full-sample counterpart of every ERP estimate reported here is recoverable. The behavioural models are unaffected, the offline filter bearing only on the EEG, and the decoding artefacts carry no tag for the switch, so a full-sample decoding run has to be kept apart from the reported one.

the onset of the critical word. Trial-by-trial artefact rejection followed, retaining on average 31.4 of the 48 presented trials per modelled cell ( $SD = 5.8$ ), a mean rejection rate of 34.5%. These retention figures are computed on the full sample, as the grand averages are, so the four mis-filtered Session 3 datasets contribute to the Session 3 means. Table 1 gives the full per-condition breakdown. Differential trial loss between the compared conditions would make a grammaticality effect partly a difference in measurement precision and only partly one in brain response, so retention is reported per condition, and the two prove effectively equal. Averaged over properties and sessions, 31.4 trials were retained on grammatical trials and 31.4 on ungrammatical trials, a difference of 0.02 trials.

For each retained trial, we computed the mean amplitude in three theoretically motivated latency windows, 200–500 ms (the LAN/early-negativity range), 300–600 ms (the N400 range) and 400–900 ms (the P600 range), at lateral and midline electrode macroregions. The windows are named by their nominal bounds throughout. On the 2 ms acquisition grid, each is inclusive of samples up to 498, 598 and 898 ms respectively. That convention is inherited from the validated importer and is noted here so that the figures and the code can be read against the prose. The lateral region comprises left- and right-hemisphere anterior, medial and posterior clusters and so carries both the hemisphere and the caudality contrasts. The midline region comprises three clusters centred on the midline and supports the caudality contrast alone. Those clusters are parasagittal, so they are not strictly midline. Besides Fz, Cz, Pz and Oz they include the pairs FC1/FC2 and CP1/CP2, which the cap places about a quarter of the head radius either side of it. Because each pair is exactly symmetric, however, the cluster mean remains centred on the midline and has no net hemispheric bias, which is why no hemisphere term is estimated there. Two further choices, the within-participant  $z$ -scoring of single-trial amplitudes and the treatment of each trial's pre-stimulus mean amplitude (–100 to 0 ms) as a covariate instead of a subtracted baseline, are modelling decisions and are set out under *Data analysis*.

### **Grammaticality-judgement task**

The grammaticality judgement was collected concurrently with the ERP recording. On each experimental trial, participants judged whether the sentence was grammatical for the property under

**Table 1***Retained Single-Trial EEG per Modelled Cell, by Grammatical Property and Session*

| Property                     | Condition     | S2   | S3   | S4   | S6   |
|------------------------------|---------------|------|------|------|------|
| Gender agreement             | Grammatical   | 31.5 | 32.1 | 32.2 | 30.2 |
| Gender agreement             | Ungrammatical | 30.6 | 32.0 | 31.4 | 30.6 |
| Differential object marking  | Grammatical   | n/a  | 32.0 | 31.7 | 30.5 |
| Differential object marking  | Ungrammatical | n/a  | 30.8 | 31.6 | 31.5 |
| Verb–object number agreement | Grammatical   | n/a  | n/a  | 31.5 | 30.6 |
| Verb–object number agreement | Ungrammatical | n/a  | n/a  | 32.1 | 31.8 |

*Note.* Values are mean trials retained of the 48 presented, averaged over language. Rows are split by grammaticality because differential trial loss across the compared conditions can bias both amplitude and decodability. The two conditions are retained almost equally here (see text). An n/a entry marks a session at which the property had not yet been introduced. Like the grand averages, the table is computed on the full sample, including the four mis-filtered Session 3 datasets that the models exclude.

test, so the trial-level accuracy (correct/incorrect) and the single-trial ERP are two measures of one and the same event, recorded within a single task. Accuracy is the sole behavioural dependent measure. Response times served only to screen trials. Judgements of 200 ms or less, or of 4,000 ms or more, were discarded as implausible, which removed 8,504 of the 42,116 otherwise valid judgement trials (20.2%), leaving 33,612 trials for analysis. The bounds were carried over from the legacy preprocessing pipeline, which recorded no rationale for them, so we disclose the screen and its cost without defending the cutoffs. Two features of the Session 4 logs removed further judgements before that screen. The runs of two participants were exported in two files whose names the extraction's file pattern does not match, and the logs of two participants were exported with a semicolon delimiter that the comma-delimited reader cannot parse. Those participants contribute no Session 4 judgements and enter the models through their other sessions only. Rows that the presentation software logged twice under one trial number were reduced to one per participant, session, property and trial, and the unlabelled filler sentences shown at Session 2 carry no property and fall outside the modelled contrast. The judgement sentences were generated combinatorially, which has a consequence for the accuracy models' random-effect structure (see *Data analysis*).

## Data analysis

All models were fitted in a fully Bayesian framework with brms (Bürkner, 2017, 2018) as a front end to the Stan No-U-Turn sampler (Carpenter et al., 2017). A Bayesian treatment is not incidental to the question here. The hypotheses concern the size and direction of effects that are expected to be small. Several of the comparisons that matter, such as whether a property's transfer effect grows across sessions, are naturally expressed as statements about a posterior. Priors, sampler settings, diagnostics and inference rules are set out in the sections that follow. The analysis and reporting code was written with the assistance of Claude Code (version 2.1, Anthropic), running primarily the Claude Opus and Claude Fable models, which drafted and revised scripts under the authors' direction and checked them against the method reported here. The authors reviewed and ran every script, and the deposited code reproduces the analyses without the tool.

### *Predictor coding and standardisation*

Every continuous predictor was  $z$ -standardised and every dichotomous predictor was  $\pm 0.5$  contrast coded before model fitting (the  $\pm 0.5$  codes are then  $z$ -standardised with the other predictors, which preserves their sign). On this common standardised scale, regression coefficients are directly comparable, and lower-order terms in interactive models retain interpretable main-effect meanings (Brauer & Curtin, 2018; Schielzeth, 2010). Standardising the inputs is also what makes a prior on a coefficient interpretable, because it fixes the scale that prior is placed on (Gelman et al., 2008). Session was recoded to consecutive integers in the order the four sessions occurred, Sessions 2, 3, 4 and 6 becoming 0–3, and then  $z$ -standardised (within property for the accuracy models, whose properties span different session ranges). The change in an effect between any two measured sessions, including across the retention interval, is then a single linear combination of the posterior draws. Besides grammaticality (grammatical = +0.5) and language (Mini-Norwegian = +0.5), stated with the design, the topographic contrasts are coded hemisphere right = +0.5, left = -0.5, and caudality anterior = +0.5, medial = 0, posterior = -0.5. A positive coefficient on a topographic interaction therefore indicates a larger effect over right or anterior sites respectively.

One further covariate enters the ERP models. It is the L1 Dominance Score, the

per-participant derived score of that name in the Language History Questionnaire's Aggregate Scores export (LHQ3: [Li et al., 2020](#)), read from that export and z-standardised with the other continuous predictors. It indexes how far the first language dominates the participant's repertoire, and it is carried in the ERP models as a control for prior language experience. The variable is named for multilingual language diversity in the extraction code, a legacy of an earlier version of the export in which the two scores sat in different columns, so the code label and the quantity modelled differ; the models use the score described here. The covariate is not carried into the accuracy models. That asymmetry has no justification, and we return to it in the Limitations.

The response side has two standardisation decisions of its own. Single-trial amplitudes were z-scored within participant. The reasoning is the one Faust et al. ([1999](#)) set out for response latencies. When individuals differ in the overall scale of a measure, and those differences are roughly linear, a group whose scale is larger will show a larger raw treatment effect, and the resulting overadditive interaction is an artefact of scale. They recommend a within-individual z-transformation as the remedy. Single-trial ERP amplitude has the same property, and the risk applies directly here because the language contrast is between participants, so a scale difference between the two groups could otherwise present as a grammaticality  $\times$  language interaction. We also entered the trial's own pre-stimulus mean amplitude (–100 to 0 ms) as a within-participant z-scored covariate. Subtraction of a baseline is the special case of this model in which the baseline coefficient is fixed at one. Estimating that coefficient lets the model absorb baseline variance that subtraction would leave in the residual, which Alday ([2019](#)) shows can raise power. It also avoids the assumption subtraction requires, that conditions do not differ systematically during the baseline interval.

***Model structure***

For each ERP cell (property  $\times$  window  $\times$  macroregion), we fitted a Gaussian model of single-trial mean amplitude:

$$\begin{aligned} \text{amplitude} \sim & \text{baseline} + \text{Gram} \times \text{Session} \times \text{Language} \\ & + \text{Gram} \times \text{Hemisphere (lateral only)} + \text{Gram} \times \text{Caudality} \\ & + \text{L1 dominance} \\ & + (1 + \text{Gram} + \text{Session} \mid \text{participant}) + (1 + \text{Gram} + \text{Session} \mid \text{item}). \end{aligned}$$

For each property, we fitted a Bernoulli (logit-link) model of judgement accuracy (Dixon, 2008; Jaeger, 2008):

$$\text{correct} \sim \text{Gram} \times \text{Session} \times \text{Language} + (1 + \text{Gram} + \text{Session} \mid \text{participant}).$$

Participants and items are crossed rather than hierarchically nested. Within a language, every participant sees every sentence frame, and every frame is seen by every participant in that language, so the ERP models carry crossed by-participant and by-item random effects (Baayen et al., 2008; Barr et al., 2013). Items are nested within language, because the two miniature languages draw on separate lexicons, so language does not vary within an item and no by-item language slope is estimable. Grammaticality and session, by contrast, vary within both grouping factors, since every frame is presented in both a grammatical and an ungrammatical version and is re-presented across sessions.

Random effects therefore follow the “keep it maximal *as justified by the design*” principle (Barr et al., 2013), so both grouping factors receive a random intercept plus grammaticality and session slopes, with correlations estimated under the shared LKJ(2) prior.

Session enters that specification as a single linear predictor, so the change in the grammaticality effect across the retention interval is read off the fitted trend rather than estimated for that interval on its own. Gender agreement and differential object marking were therefore also refitted with a binary indicator for the retention session, interacted with grammaticality and language and added to the linear session term. The linear term then carries the training trend and

the indicator the deviation at Session 6. The indicator is coded +0.5 at Session 6 and −0.5 elsewhere and, unlike the other predictors, is not standardised, so its coefficient is the deviation at Session 6 on the response scale. The refits are otherwise identical to the models above and were run under both random-effect structures. Verb–object number agreement admits no such refit. It is sampled at the two bracketing sessions only, so an indicator for the later of them would be collinear with the linear term. Where the two derivations disagree, the freed estimate is the one that bears on consolidation, and both are reported (Table B1, Table B2).

The case for retaining the by-item grammaticality slope, and the power-based argument against maximal structures that it was weighed against (Barr et al., 2013; Matuschek et al., 2017), are set out in Appendix A. The Results report the whole grid refitted without that slope, so a reader can see which conclusions depend on it. The accuracy models omit a by-item effect by design. The judgement sentences are generated combinatorially. In Session 4, the 11,541 judgement trials of 52 participants presented 1,720 distinct sentences, each on average 6.7 times across the sample. Item variance is therefore thinly sampled per sentence, and no by-item term is carried. Its variance is absorbed into the residual, which leaves the accuracy estimates slightly less precise and is noted under *Limitations* (Barr et al., 2013).

### ***Priors***

We placed informative priors derived from the morphosyntactic-ERP and behavioural literature, in place of the common weakly informative default, following a selective, sign-informed and magnitude-regularising strategy (Schad et al., 2021). Because the response and all predictors are standardised, each coefficient is expressed in standard-deviation units (within participant, for the ERP models). A prior *mean* can therefore encode the direction of a well-established effect, while its *standard deviation* stays wide enough to let the data dominate any well-identified effect (Gelman et al., 2008).

The one firmly established directional effect is the P600: words that violate the syntactic structure of a sentence elicit a late positivity, larger for ungrammatical than for grammatical sentences (Osterhout & Holcomb, 1992). For morphosyntactic violations specifically, that positivity

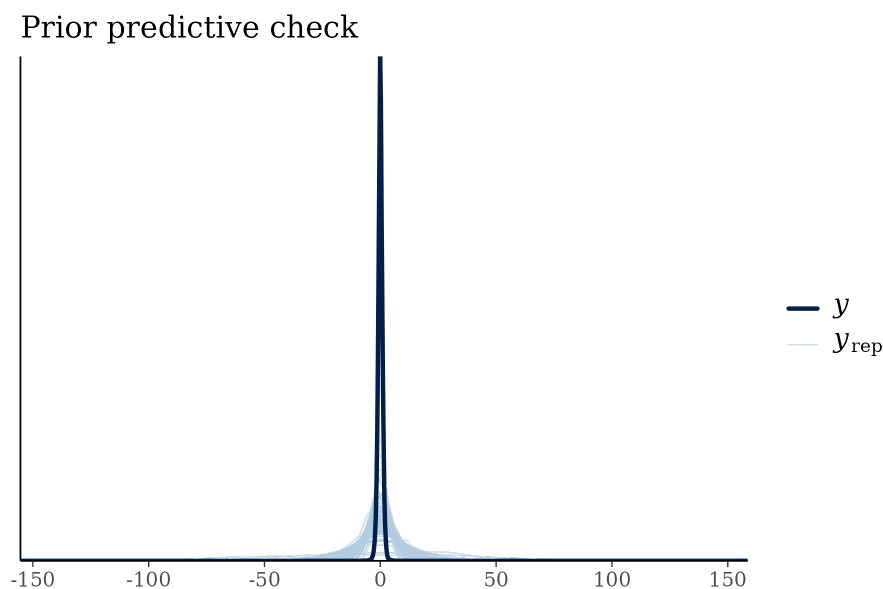
is posterior and centroparietal, and it is obtained in native speakers (Chen et al., 2007; Mueller et al., 2005, 2007) and, once some proficiency has been reached, in artificial-language learners (Morgan-Short et al., 2010; Morgan-Short, Steinhauer, et al., 2012). A P600 also arises to structural violations in implicit artificial-grammar learning, where the sequences carry no morphology at all (Silva et al., 2017). Because grammaticality is coded grammatical = +0.5, this implies a *negative* prior mean on the grammaticality slope in the P600 window, carried more mildly into the 300–600 ms window for the two agreement properties, whose late signature is a positivity rather than an N400. For differential object marking, the specification differs in both windows, being widened in the later window and left zero-centred in the earlier one, because case-marking violations are configuration-dependent and elicit a P600, an N400 or a biphasic response according to animacy and markedness (Frisch & Schlesewsky, 2001; Mueller et al., 2007; Nieuwland et al., 2013). As the P600 is posterior-maximal (Mueller et al., 2007), the grammaticality  $\times$  caudality interaction received a positive prior mean in each window in which the grammaticality prior carried a sign. The early (200–500 ms) anterior-negativity window and the hemisphere interaction were left zero-centred, because the LAN is small and present in only roughly half of participants and items, so its sign is not safely predictable (Caffarra et al., 2019). The grammaticality  $\times$  session  $\times$  language transfer interaction, the hypothesis under test, was kept zero-centred, because the second-language signature of these effects is qualitatively contested (Chen et al., 2007; Mueller et al., 2009) and proficiency-dependent (Caffarra et al., 2015). The priors thus inform the model where the evidence is strong without biasing the effects we set out to estimate.

For the accuracy models, artificial-language learners judge grammaticality well above chance and improve across successive test blocks even under mere exposure, without explicit instruction or feedback (Mueller et al., 2009), motivating a positive intercept and a positive session slope. Random-effect standard deviations, and the residual standard deviation of the ERP models, received a generous half-Student-*t* prior, a weakly informative default for variance parameters in hierarchical models (Gelman, 2006). Table C1 in Appendix C gives the distribution placed on every parameter. A generous prior is warranted here because between-participant variation in the

morphosyntactic ERP response is not incidental noise. Tanner and Van Hell (2014) found that although the grand mean showed the usual biphasic LAN–P600 pattern, individual responses varied systematically along a continuum between negativity-dominant and positivity-dominant, so participants differ in the form of the response and not only in its size. Modelling that variation as a random effect over single trials, instead of averaging across it, is the approach Volpert-Esmond et al. (2021) set out for ERP data. Random-effect correlation matrices received an LKJ(2) prior (Lewandowski et al., 2009). The literature does not supply a defensible microvolt effect size for these components (even systematic resampling finds the magnitude of the P600 to vary widely across samples, Kim et al. (2024)). Prior means therefore encode only sign and a regularised magnitude, and their influence is bounded by deliberately wide standard deviations. Prior predictive simulation confirmed that the priors imply amplitude and accuracy distributions that encompass the observed data (Figure 4). Every model in the grid, under both random-effect structures, and every accuracy model was also refitted on the same data, as a sensitivity analysis, under the former weakly informative priors of  $\mathcal{N}(0, 1)$  and  $\mathcal{N}(0, 1.5)$ , and the estimates were essentially unchanged. The retention-free refits were fitted under the informative priors only. The largest shift in any grammaticality term was 0.0008 on the within-participant standardised scale for the ERP models and 0.049 in log-odds for the accuracy models. No grammaticality term’s credible interval changed width by more than 2.6% in the ERP models or 2.8% in the accuracy models. Taken over every reported parameter, the largest width change was 7.3%. The informative priors therefore regularise without driving the conclusions (Depaoli & van de Schoot, 2017; Schad et al., 2021; van de Schoot et al., 2021). The smallest of the credible transfer effects (differential object marking, late lateral window) is unchanged under the weak priors ( $b = -0.011$ , 95% CrI  $[-0.020, -0.003]$ ,  $p_d = .997$ ), so its credibility does not depend on them.

### ***Sampler settings and convergence***

Each model was run with four chains of 10,000 iterations, the first 2,000 of each discarded as warm-up, with `adapt_delta = 0.95` and `max_treedepth = 12` to accommodate the challenging geometry of maximal multilevel posteriors. Convergence was assessed with the

**Figure 4***Prior Predictive Check for a Representative Model*

*Note.* The model is subject–verb gender agreement, 400–900 ms midline. Light lines are datasets simulated from the priors alone, with the likelihood switched off, and the dark line is the observed within-participant z-scored single-trial amplitudes. The prior predictive distribution comfortably encompasses the observed data, so the priors are not in conflict with it. The horizontal axis spans the full range of the prior draws, which reach far beyond the observed range. That is the intended behaviour of wide priors on a standardised scale, and it is why the observed density appears compressed near zero. The check therefore establishes that the priors generate data of a plausible order of magnitude, not that they are tightly calibrated to the observed distribution.

rank-normalised split- $\hat{R}$  and bulk/tail effective sample sizes (Vehtari et al., 2021), requiring  $\hat{R} < 1.01$ , bulk- and tail-ESS  $> 400$ , and no divergent transitions. Model fit was checked with posterior predictive checks (Gabry et al., 2019). These two thresholds do different jobs. An effective sample size of 400 (at least 100 per chain) is the point at which the  $\hat{R}$  and ESS diagnostics are themselves reliable (Vehtari et al., 2021). The stability of a 95% interval limit is a more demanding requirement, because extreme quantiles are estimated far less efficiently than a posterior mean (Kruschke, 2021). The realised sampling is substantially above these thresholds (the lowest bulk- and tail-ESS over all parameters of all models are reported in Table D1 in Appendix D). Coefficients whose intervals lie close to zero should nonetheless be treated as borderline.

### *Inference*

We summarise each effect by its posterior median, 95% credible interval (CrI) and probability of direction ( $p_d$ , the posterior probability that the effect has its median's sign), an estimation-first approach that avoids dichotomous null-hypothesis testing (Makowski et al., 2019). We describe an effect as *credible* when its 95% credible interval excludes zero, and otherwise read  $p_d$  as a continuous index of directional certainty (Kruschke & Liddell, 2018). All credible intervals are equal-tailed quantile intervals bounded by the 2.5th and 97.5th percentiles of the posterior draws. For a skewed posterior, an equal-tailed and a highest-density interval can differ in whether they contain zero, so the choice is stated (Kruschke, 2021). Within each model, hierarchical shrinkage across participants and items already regularises the coefficient estimates (Gelman et al., 2012; Kruschke & Liddell, 2018). The eighteen cells are fitted separately and are not pooled towards one another, so that shrinkage is not itself an argument about the grid. The grid (3 properties  $\times$  3 a priori windows  $\times$  2 macroregions) is fixed before fitting, and every cell is reported without selection (Figure 5 reports all 18 cells for the key terms, and Table B1 the derived retention contrasts). No correction across cells is applied, and no cell is screened for significance before being reported. Where a conclusion does rest on particular cells crossing the credibility threshold (the transfer interactions are credible in 3 of 18 cells), we treat it as provisional on that basis (see *Strength of evidence* in the Discussion). We do not claim the grid is immune to selection effects (Rubin, 2021). The derived retention effect (Session 4  $\rightarrow$  Session 6) and its language-scoped contrasts are posterior quantities computed from the same models, summarised and disclosed in full the same way (Table B1). For gender agreement and differential object marking, they are also re-estimated in refits that free the Session-6 step from the training trend, described under *Model structure* (Table B2).

### *Multivariate decoding of grammaticality*

The Bayesian models above summarise each electrode cluster's mean amplitude separately, which could miss a grammaticality signal that is distributed across the scalp too weakly to move any single cluster's mean but is nonetheless jointly decodable from the whole-scalp pattern (Haxby et al., 2001). As a complementary, classical (non-Bayesian) analysis, we decoded Grammatical from

Ungrammatical trials directly from the single-trial amplitude pattern across all EEG sensors at every time sample (baseline-corrected, downsampled to a 4 ms grid). The classifier was a regularised linear discriminant analysis under leave-one-participant-out cross-validation, a time-resolved multivariate pattern analysis (MVPA) approach for EEG/MEG ([Grootswagers et al., 2017](#)). Leave-one-participant-out is the generalisation test this question calls for, since what is at stake is whether the pattern transfers to an unseen learner. Varoquaux et al. (2017) show that leave-one-out schemes give more variable accuracy estimates than repeated random splits, so we read the AUC time-course below as a profile whose individual point estimates carry limited precision. Decoding performance is summarised as the area under the ROC curve (AUC) at each time point. Chance-level calibration was established with a permutation test that shuffles the grammaticality labels once per permutation, holding the shuffle constant across the whole time-course so that the null preserves each trial's temporal autocorrelation. Multiple comparisons across time points were corrected with cluster-based permutation inference ([Maris & Oostenveld, 2007](#)). Clusters were formed from temporally contiguous samples whose AUC exceeded the 95th percentile of the pooled permutation null, the cluster statistic was the summed excess of AUC over that threshold, and each cluster was assessed against the distribution of the largest cluster statistic across permutations. A second correction layer controls the false-discovery rate (FDR) across clusters, with the family defined per property. Benjamini–Hochberg correction ( $q < .05$ ) is applied across the clusters of a property's pooled overall time-course, and each property's family is corrected separately. The session-resolved and session-by-language breakdowns carry only their own cluster-level  $p$  values. [Appendix A](#) sets out why the family is defined this way. Grammatical and ungrammatical trial counts differ across participants and decline across sessions, so we balanced the classes by undersampling the majority class within each participant of the training fold. The discriminant's pooled within-class covariance was ridged by one thousandth of its mean diagonal for numerical stability. The AUC is in any case insensitive to class prevalence, so neither the classifier nor the metric can exploit a prevalence shortcut, and chance was set by the empirical within-participant permutation null ([Combrisson & Jerbi, 2015](#)). The permutation null comprised

1,000 shufflings, so the smallest attainable cluster  $p$  is .0010. This bounds the resolution of near-threshold cluster  $p$ -values, and the surviving clusters are reported with that granularity in mind (T. E. Nichols & Holmes, 2002; Phipson & Smyth, 2010). One qualification belongs with that description. The confirmatory time-courses reported here were corrected under exactly the family just defined, and at the full permutation null. The exploratory breakdowns for gender agreement, which enter no family, are still those of an earlier run at 200 shufflings, so their cluster-level  $p$  values resolve less finely than the confirmatory ones. A rerun that brings them to the same null is under way, and by construction it cannot alter a confirmatory decision. The decoder is given the EEG sensors alone. The amplifier's built-in accelerometer channels are excluded, as is any channel not exported for every session a property was recorded in, so that each property's pattern is brain-derived and no sensor is present for one session's trials but structurally absent for another's. Beyond the overall time-course, five further analyses were run. Decoding separately by session, and by session  $\times$  mini-language, gives a multivariate analogue of the ERP models' grammaticality  $\times$  session and  $\times$  language terms. A cross-language generalisation analysis trains the decoder on one mini-language and tests it on the other. A cross-property generalisation analysis trains on one grammatical property's Grammatical/Ungrammatical contrast and tests on another. Finally, temporal generalisation trains a decoder at every time point and tests it at every other, on a grid coarsened by a factor of two on both axes to keep the computation tractable. It characterises whether the same neural code is stable across the epoch or evolves (King & Dehaene, 2014). The two generalisation analyses carry no permutation null and are reported as descriptive comparisons. Cross-validating strictly out-of-participant throughout guards against the circular, non-independent analysis that would otherwise inflate apparent decoding accuracy (Kriegeskorte et al., 2009).

### **Transparency, preregistration and analysis status**

The study design, artificial-language materials and session protocol were preregistered before data collection (<https://osf.io/tjr54>). The preregistration specified the design, the miniature-language materials, the six-session schedule and the EEG acquisition, but it did not commit to a statistical analysis plan. The Bayesian multilevel models, the literature-derived

informative priors, the latency windows and macroregions, the Session 4 → 6 retention contrasts and the multivariate decoding analyses reported here were therefore specified after the data were collected. None was preregistered, so all are exploratory in the registered sense, and we report them as estimation-oriented analyses, since none tests a preregistered hypothesis (Nosek et al., 2018; Wagenmakers et al., 2012). Within that frame, we distinguish *primary* from *secondary* analyses. The primary analyses, fixed before any model was fitted, comprise the ERP grid of properties, windows, macroregions and priors together with the pooled, all-sessions decoding time-course for each property. The secondary analyses are the session-resolved and session-by-language decoding breakdowns, the cross-language and cross-property generalisation analyses, and the temporal-generalisation matrix. What the paper reports of them varies, and we say which is which. The cross-language and cross-property peaks are quoted in the Results. The session-resolved and session-by-language breakdowns and the temporal-generalisation matrices are not summarised numerically here and are available in the deposited decoding artefacts. Where a secondary analysis carries a cluster-level *p* value, that value is family-wise-error-controlled over time in the usual way, but none of them is treated as a test of the paper’s claims. Where the decoding analyses speak of a “confirmatory family”, the term names the false-discovery-rate correction family (the clusters of each property’s pooled overall time-course), not preregistered confirmation. On a minor point of adherence, a third, ancillary violation type was recorded per property but, as noted above, is not analysed here.

Two decisions were made after inspecting results, and we label them as such. First, peak decoding accuracies are quoted on the post-stimulus axis, a restriction adopted once the pre-stimulus interval was found not to be at chance for differential object marking. The pre-stimulus interval is retained and reported alongside as a negative control. Second, the temporal-generalisation and cross-property analyses were reclassified as descriptive after it became clear that no null distribution had been computed for them. Neither decision creates a new test, and both move the reported result in the conservative direction. One analysis is not yet uniform, and we flag it here as well as where it is reported. The confirmatory decoding time-courses are final: they

are corrected under the family the Method describes and rest on the full permutation null. The session-resolved breakdowns for gender agreement are still those of an earlier, coarser run, and a rerun that brings them to the same null is in progress. Because those breakdowns enter no correction family, the rerun cannot alter a confirmatory decision, in either direction.

The preregistration planned data collection at two sites, in Norway and Spain. The present analyses concern only the Norwegian cohort (first-language Norwegian, second-language English learners of Mini-English or Mini-Norwegian). The Spanish-site data are not analysed here. The findings therefore speak to first-language-Norwegian learners, and generalisation to the other registered populations is an open question.

Following the preregistered protocol, in each grammar session, participants had to score above 80% on the post-training behavioural test (with a second attempt allowed) before proceeding to the EEG experiment. This performance threshold is both an inclusion criterion and a performance-conditional missingness mechanism, since it conditions which participant-sessions contribute EEG and behaviour. Its effect is therefore folded into the attendance counts in Figure 3, which record who contributed data at each stage but not why. Per-session gate outcomes are not reported. They can only be reconstructed from the terminal pass flag in the training logs, and that reconstruction does not reconcile with the attendance counts at Session 4, so we do not quote numbers from it.

## Results

All 102 fitted models met the convergence criteria. Taken over every parameter of every model, the largest rank-normalised  $\hat{R}$  was 1.007, the smallest bulk and tail effective sample sizes were 593 and 1,128, and no model produced a divergent transition. Per-model values are given in Table D1.

### ERP signatures of grammatical processing and their development

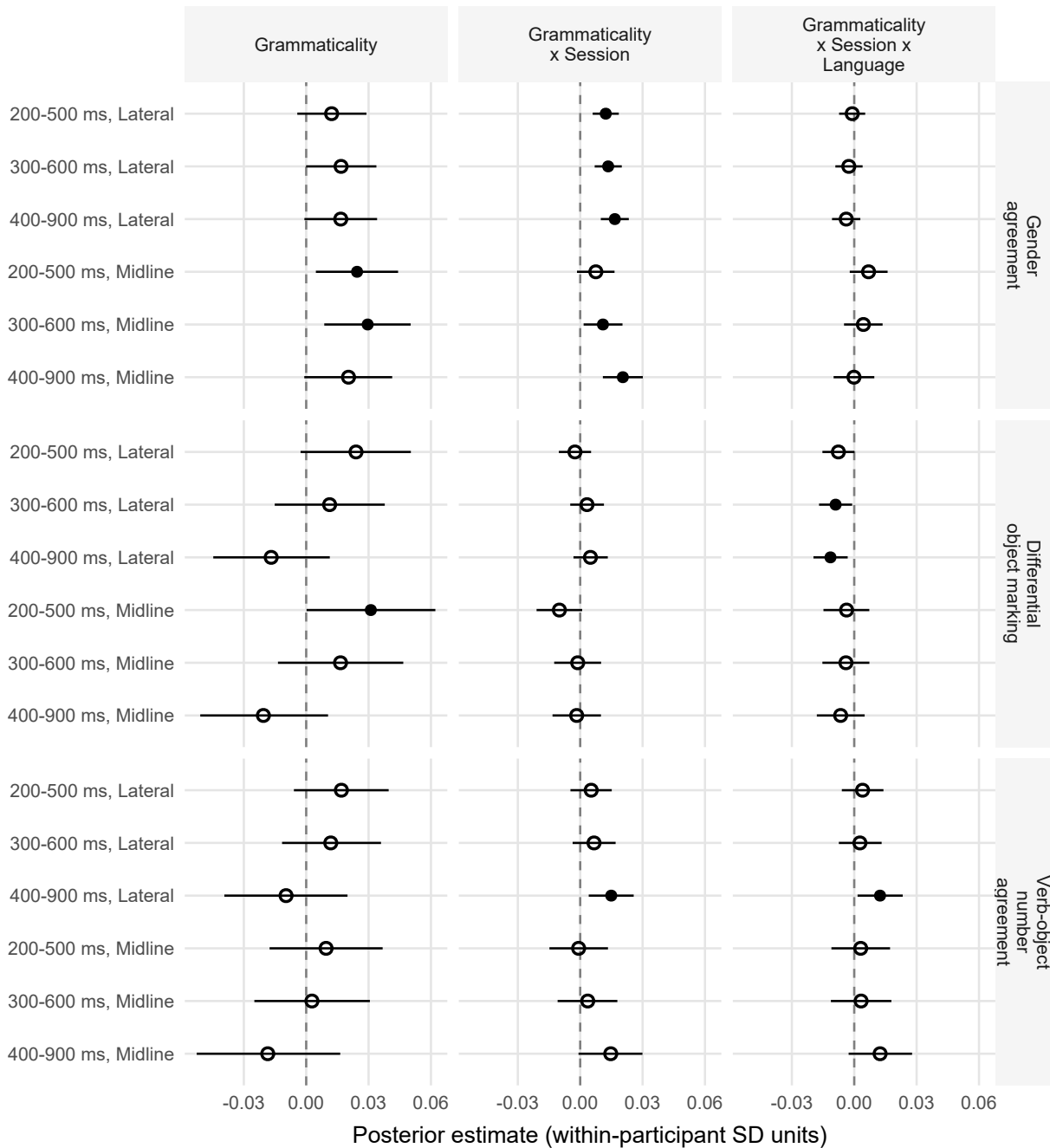
We assess, for each property, whether it elicited a grammaticality effect, whether that effect sharpened across sessions (the grammaticality  $\times$  session interaction), and whether that development was modulated by language (the grammaticality  $\times$  session  $\times$  language interaction). The overall

pattern is property-specific, with gender agreement showing the clearest and most broadly distributed learning effect, credible in 5 of 6 cells; verb–object number agreement a late, language-conditioned effect; and differential object marking a transient early grammaticality effect with a late, language-conditioned modulation (Figure 5). The underlying condition averages are shown as grand-average waveforms in Figure 6. Because those overlaid waveforms are dominated by the shared evoked response, the difference waves that isolate the grammaticality effect are given in Appendix E (Figure E1).

One coding fact governs how every coefficient below should be read. A positive grammaticality coefficient means that grammatical trials were more positive than ungrammatical ones, because grammaticality is coded grammatical = +0.5. Violations therefore appear as a relative negativity. The canonical P600 would produce a negative coefficient. Every credible grammaticality effect below is positive, so the dominant signature in these learners is a violation-related negativity where our priors anticipated a late positivity, a point we return to in the Discussion. Results are organised by property.

### ***Subject–verb gender agreement***

Gender agreement showed the clearest profile. The grammaticality effect was positive in direction in every window and macroregion ( $p_d$  from .93 to .997; Figure 5) and credible in 2 of 6 cells, both of them on the midline. Laterally, the intervals include zero throughout. The effect also sharpened across sessions, and that development is the better supported of the two. The grammaticality  $\times$  session interaction was credible from the early window (lateral 200–500 ms,  $b = 0.012$ , 95% CrI [0.006, 0.019],  $p_d > .999$ ; midline 300–600 ms,  $b = 0.011$ , 95% CrI [0.002, 0.020],  $p_d = .99$ ) through the late window (midline 400–900 ms,  $b = 0.021$ , 95% CrI [0.011, 0.030],  $p_d > .999$ ). Source-language transfer in the ERP signal was not credible, however. The grammaticality  $\times$  session  $\times$  language interaction sat near zero in the late window ( $b = -0.0001$ , 95% CrI [-0.0099, 0.0097],  $p_d = .51$ ), with only an early-window hint (midline 200–500 ms,  $b = 0.007$ , 95% CrI [-0.002, 0.016],  $p_d = .93$ ).

**Figure 5***Posterior Medians and 95% Credible Intervals for the Key Standardised Effects Across ERP Cells*

*Note.* Cells are arranged by grammatical property (rows) and term (columns), with each cell's window and macrorregion on the vertical axis. All panels share a common x axis, so effect magnitudes are directly comparable across cells and terms. Filled points mark effects whose 95% interval excludes zero; open points do not. Estimates are in within-participant SD units.

**Figure 6***Grand-Average ERPs per Property at Two Representative Sites*

*Note.* Waveforms are shown at a midline posterior region (parieto-occipital, spanning the P600) and a left-anterior region (spanning the LAN/early negativity), grammatical against ungrammatical. Each line is the mean over all retained trials of all contributing participants, so participants with more retained trials contribute more to the mean. Positive voltage is plotted upward. Time zero is critical-word onset, and the pre-stimulus baseline (–100 to 0 ms) is shaded. The vertical axis is scaled separately for the two sites. The waveforms are not baseline-corrected (the single-trial models use each trial's own pre-stimulus baseline). The three a priori analysis windows (200–500, 300–600 and 400–900 ms) are shaded. They overlap by design, because the underlying components do, so the depth of shading reflects window overlap and does not mark out any one window. These are descriptive condition averages with no uncertainty band, and the inferential statements come from the models in the forest plot. Unlike those models, they are computed on the full sample, including the four mis-filtered Session-3 datasets described in the Method.

***Differential object marking***

Differential object marking showed an early, transient grammaticality effect (grammatical trials more positive, a relative violation negativity), and no developing late effect. The effect was credible in the early midline window alone (Figure 5; 200–500 ms,  $b = 0.0311$ , 95% CrI [0.0002, 0.0621],  $p_d = .98$ ), with its lateral counterpart positive but not credible ( $b = 0.024$ , 95% CrI [−0.003, 0.050],  $p_d = .96$ ), and was, if anything, slightly negative by the late window (midline 400–900 ms,  $b = -0.021$ , 95% CrI [−0.051, 0.011],  $p_d = .91$ ). One caveat attaches to this early effect specifically. Because the violation omits the particle *fi*, the two conditions differ physically in the region preceding the critical word. At the roughly 505 ms word-onset asynchrony, the response to that region overlaps the 200–500 ms window of the critical epoch. The same low-level difference that renders the pre-stimulus decoding baseline non-chance (see below) could therefore contribute here, so we read the early effect as a violation-related response with a possible stimulus-driven component. Its development was not credible ( $b = -0.002$ , 95% CrI [−0.013, 0.010],  $p_d = .61$ ), but the grammaticality  $\times$  session  $\times$  language transfer interaction was credible in the two later lateral windows (300–600 ms,  $b = -0.009$ , 95% CrI [−0.017, −0.001],  $p_d = .99$ ; 400–900 ms,  $b = -0.011$ , 95% CrI [−0.020, −0.003],  $p_d = .997$ ). The effect carried the same negative sign (a development smaller for the Mini-Norwegian group, given the coding above) in all six cells and did not reach credibility on the midline (400–900 ms,  $b = -0.007$ , 95% CrI [−0.018, 0.005],  $p_d = .86$ ), so it forms a coherent lateral, late-window pattern that spans several cells. Its sign is opposite to the verb–object transfer effect reported next, so the two properties diverge in direction as well as in magnitude.

***Verb–object number agreement***

Verb–object number agreement, the latest-introduced property, developed only in the late window (Figure 5), and there it bore a source-language signal. The grammaticality  $\times$  session interaction was credible laterally (400–900 ms,  $b = 0.015$ , 95% CrI [0.004, 0.026],  $p_d = .997$ ), as was the grammaticality  $\times$  session  $\times$  language transfer interaction ( $b = 0.012$ , 95% CrI [0.002, 0.023],  $p_d = .99$ ). On the midline, the same interaction was not credible ( $b = 0.012$ , 95%

CrI  $[-0.003, 0.028]$ ,  $p_d = .95$ ). The lateral effect is the only positive one among the 3 of 18 cells in which a transfer interaction is credible, and every one of those cells is lateral. Midline intervals are wider throughout the grid because the three midline clusters supply half as many observations as the six lateral ones.

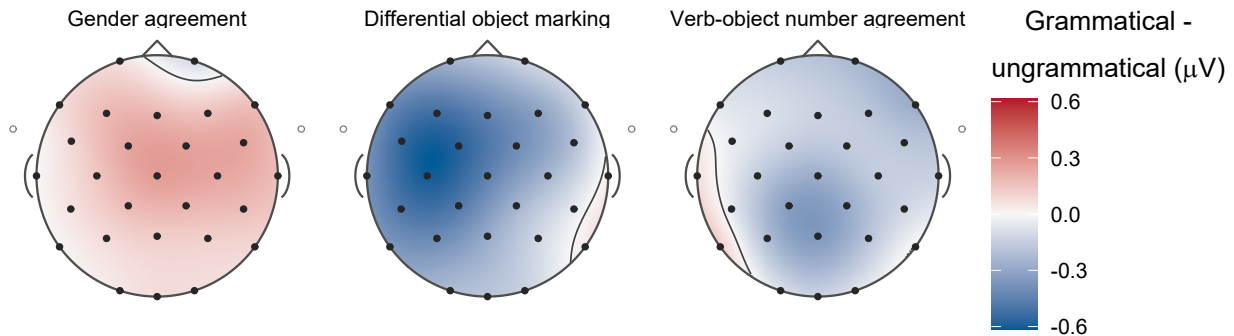
One reading caveat applies to this property alone. Verb–object number agreement is measured at Sessions 4 and 6 only, and the interval between them is the unrehearsed retention gap, so for this property the session term is the change across that gap and contains no further exposure. Its grammaticality  $\times$  session and three-way terms are therefore offline-interval changes, and the *learning* and *transfer* glosses that the operationalisation gives those terms for the other two properties do not carry over unaltered here. We read them accordingly throughout.

### ***Topography***

Grammaticality interacted with hemisphere for differential object marking (200–500 ms,  $b = 0.019$ , 95% CrI  $[0.012, 0.026]$ ,  $p_d > .999$ ), the effect being larger over the right hemisphere (hemisphere is coded right = +0.5). The priors were neutral on this asymmetry (zero-centred, because the one lateralised component is too heterogeneous to sign a priori), so nothing about it was predicted, and no by-participant slope was estimated for it. The grammaticality  $\times$  caudality interaction, by contrast, was not credible in the late window for any property (all  $p_d$  between .52 and .87). Raw voltage varies with anterior–posterior position, as it does across any scalp recording, but the grammaticality effect itself was no more pronounced posteriorly than anteriorly, so the topography does not confirm a canonical posterior-maximal P600 distribution (Figure 7).

### ***Dependence on the by-item random-effect structure***

Everything above comes from the maximal structure, with crossed by-participant and by-item random effects each carrying a grammaticality slope. Refitting the whole grid without the by-item grammaticality slope leaves every transfer conclusion unchanged (3 of 18 credible cells against 2 of 18) but narrows the grammaticality main effect's intervals by about 1.13 times, so the breadth of that effect across cells is structure-dependent while its existence and direction hold under both. The comparison is set out in full in [Appendix A](#).

**Figure 7***Scalp Topography of the Grammaticality Effect in the 400–900 ms Window, per Property*

*Note.* The topography is the grammatical minus ungrammatical mean amplitude, averaged over all retained trials and so weighted by each participant's retained-trial count, with no baseline correction. As with the waveforms, the average is taken over the full sample, including the four mis-filtered Session-3 datasets that the models exclude. The field is interpolated by thin-plate spline across the 30 electrodes assigned to the analysis regions (dots; nose at the top) and masked to the head, so no value is shown for scalp the montage did not sample. Sensor positions are the study's own recorded montage, read from the BrainVision headers. The outline marks the equator, and ears are drawn for orientation. The two low-temporal sites that sit below the equator (FT9/FT10) therefore fall outside the outline and are drawn as open circles. They inform the interpolation at the rim but no value is plotted beyond the equator. Warmer values indicate a larger grammatical-trial amplitude. The colour scale is symmetric about zero and shared across properties, and the grey line is the zero contour, so the sign of a region can be read without the colour. The distribution is diffuse and not posterior-maximal, consistent with the model-based grammaticality-by-caudality interaction not being credible for any property in this window (see main text).

### **Retention (Session 4 to Session 6)**

The retention contrast quantifies the change in the grammaticality effect between Sessions 4 and 6, positive values indicating an effect that strengthened over the interval and values near zero an effect that held steady. Its derivation bounds what it can show. Session enters the models as a single linear predictor, so the contrast is the per-session change in the grammaticality effect rescaled to the width of the retention step. Every session in the series informs that estimate, including those outside the retention interval. For gender agreement, sampled from Session 2, and for differential object marking, sampled from Session 3, the contrast describes the fitted trajectory evaluated over the retention interval. It cannot by itself separate an offline gain specific to that interval from continued

improvement at the rate the training sessions established. Verb–object number agreement is the exception, being present at Sessions 4 and 6 only, so for that property the linear session term is precisely the change across those two sessions and the contrast is estimated directly from them. Table B1, in Appendix B, reports every contrast overall, within each language and, where derived, as the between-language difference; the text below reports the ones that carry the argument.

For gender agreement and differential object marking, that limitation is addressed directly. Both properties were refitted with an indicator for the retention session, interacted with grammaticality and language, so that the linear term carries the training trend and the indicator the deviation at Session 6. Table B2, in the same appendix, reports those freed contrasts. They answer a different question from Table B1, and where the two disagree, the freed estimate is the one that bears on consolidation. Verb–object number agreement admits no such refit, for the reason given under *Model structure*.

The retention contrasts reveal a property- and language-specific pattern. The gender-agreement grammaticality effect strengthened across the interval, credibly and in both languages (midline 400–900 ms,  $\Delta = 0.019$ , 95% CrI [0.010, 0.028],  $p_d > .999$ ). Verb–object number agreement strengthened overall (lateral 400–900 ms,  $\Delta = 0.031$ , 95% CrI [0.008, 0.053],  $p_d = .997$ ), but that gain was carried by the Mini-Norwegian group ( $\Delta = 0.055$ , 95% CrI [0.024, 0.087],  $p_d > .999$ ; Mini-English  $\Delta = 0.004$ , 95% CrI [−0.027, 0.036],  $p_d = .61$ ), and the difference between the languages is itself credible ( $\Delta = 0.051$ , 95% CrI [0.007, 0.095],  $p_d = .99$ ). Differential object marking showed no credible overall ERP retention in any cell, but its grammaticality effect strengthened within Mini-English in the two later lateral windows (300–600 ms,  $\Delta = 0.016$ , 95% CrI [0.001, 0.030],  $p_d = .99$ ; 400–900 ms,  $\Delta = 0.021$ , 95% CrI [0.007, 0.036],  $p_d = .998$ ), with the Mini-Norwegian contrasts not credible in either.

### Judgement accuracy

Overall judgement accuracy fell across sessions, credibly and substantially, for every property (gender agreement  $b = -0.53$ , 95% CrI [−0.72, −0.34],  $p_d > .999$ ; differential object marking  $b = -1.20$ , 95% CrI [−1.58, −0.81],  $p_d > .999$ ; verb–object number agreement

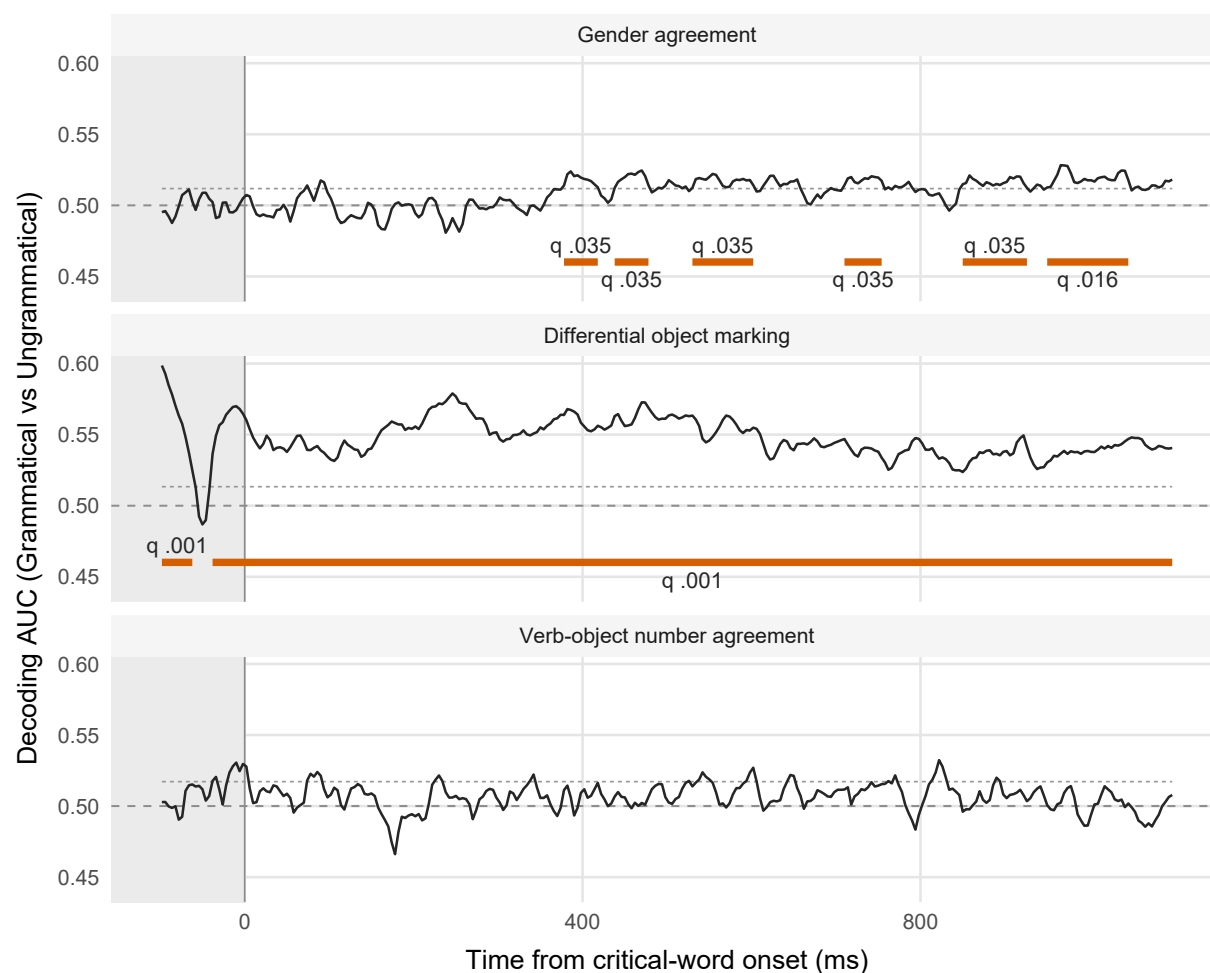
$b = -0.65$ , 95% CrI  $[-0.98, -0.32]$ ,  $p_d > .999$ ). This runs against the direction we had assumed in setting a positive prior on that term. What fell was the *level* of performance, while the ability to discriminate held. The grammaticality effect, which is what indexes knowledge of the property, is credibly positive for differential object marking ( $b = 0.87$ , 95% CrI  $[0.46, 1.29]$ ,  $p_d > .999$ ) and verb–object number agreement ( $b = 0.80$ , 95% CrI  $[0.45, 1.15]$ ,  $p_d > .999$ ). For gender agreement, the average effect is small ( $b = 0.11$ , 95% CrI  $[-0.18, 0.40]$ ,  $p_d = .78$ ), but it grows credibly across sessions (see below). Discrimination is therefore present, and improving, where the level falls. Two features of the design bear on the decline. Properties accumulate across sessions, so a judgement made at Session 4 is made against a grammar with three manipulated properties instead of one. The final session, moreover, follows a months-long interval without exposure, over which some forgetting is expected. We cannot separate these from a general drift in response criterion, because the design has no condition in which task demands are held constant while sessions advance. The decline is therefore treated as background against which the discrimination measures are read, and we draw no conclusion about learning from it.

Against that background, accuracy converged with the ERP development for gender agreement and diverged for differential object marking. For gender agreement, the learning effect was credible (grammaticality  $\times$  session,  $b = 0.21$ , 95% CrI  $[0.13, 0.28]$ ,  $p_d > .999$ ), as was a language modulation of it ( $b = 0.16$ , 95% CrI  $[0.09, 0.24]$ ,  $p_d > .999$ ), advantaging Mini-Norwegian. The gender effect also strengthened at retention, driven by that group (within Mini-Norwegian  $\Delta = 0.34$ , 95% CrI  $[0.26, 0.43]$ ,  $p_d > .999$ ; Mini-English  $\Delta = 0.04$ , 95% CrI  $[-0.07, 0.14]$ ,  $p_d = .76$ ). Differential object marking elicited a large grammaticality effect ( $b = 0.87$ , 95% CrI  $[0.46, 1.29]$ ,  $p_d > .999$ ) that nonetheless declined across the retention interval in both groups (overall  $\Delta = -0.31$ , 95% CrI  $[-0.41, -0.21]$ ,  $p_d > .999$ ). Verb–object number agreement showed a strong grammaticality effect ( $b = 0.80$ , 95% CrI  $[0.45, 1.15]$ ,  $p_d > .999$ ) with little net change at retention ( $\Delta = -0.11$ , 95% CrI  $[-0.27, 0.06]$ ,  $p_d = .89$ ).

### Multivariate decoding of grammaticality

We decoded Grammatical from Ungrammatical trials at each time sample (leave-one-participant-out AUC; Figure 8). Error was controlled across time within each analysis by cluster-based permutation and, within each property, across the clusters of its pooled overall time-course by Benjamini–Hochberg FDR (the confirmatory correction family; see *Transparency, preregistration and analysis status*). A significant cluster licenses only that the grammaticality time-course as a whole departs from chance. The temporal boundaries of a surviving cluster are not themselves inferential claims about the onset, offset or duration of the effect (Sassenhagen & Draschkow, 2019), and we do not read them as such. One qualification attaches to this section. The confirmatory cluster-level decisions are final. The session-resolved breakdowns for gender agreement still rest on the earlier, coarser permutation null, so their cluster-level  $p$  values resolve less finely, and the rerun that will bring them to the same null is still running.

The three properties differed here too, though the ordering of their raw separability has to be read against the pre-stimulus baseline. Differential object marking, whose univariate signature was the weakest and shortest-lived, was by far the most separable, though not necessarily by a violation response: its overall time-course carried two clusters surviving FDR correction, peaking post-stimulus at AUC 0.58 (at 246 ms; Figure 8). The pre-stimulus baseline, which should sit at chance, did so for the other properties but not for differential object marking, whose pre-stimulus AUC averaged 0.55 with 20 samples surviving FDR before the critical word. Its highest AUC anywhere in the epoch falls in that baseline (0.60), above its post-stimulus peak. Part of the differential-object-marking separability therefore precedes the stimulus and may reflect a low-level difference between conditions, for instance the physical presence of the object marker in the preceding region, and need not reflect a violation response. Peak AUCs are quoted on the post-stimulus axis for this reason. The extents of the surviving clusters make the limit concrete: one lies wholly before the critical word (-98 to -62 ms) and the other begins just before onset and spans the epoch (-38 to 1098 ms). The cluster test cannot attribute any part of the surviving separability specifically to a violation response. What it establishes is that the conditions are separable

**Figure 8***Time-Resolved Multivariate Decoding of Grammaticality per Property*

*Note.* Lines are leave-one-participant-out AUC for the Grammatical against Ungrammatical contrast (overall analysis, all sessions pooled). The dashed line is chance (0.5). The dotted line is the cluster-forming threshold used by the permutation test (the 95th percentile of the pooled within-participant null). Horizontal orange bars mark the temporal extent of clusters in the pooled overall time-course that survive Benjamini–Hochberg FDR correction ( $q < .05$ ) across that property’s overall clusters (each property’s family corrected separately), annotated with the cluster-level  $q$ , the Benjamini–Hochberg adjusted cluster-level  $p$  value. The bars denote cluster membership, not sample-by-sample significance. The session-resolved breakdowns are secondary and carry only their own cluster-level  $p$  values (see Transparency, preregistration and analysis status). The AUC line is a point estimate. The pipeline records no per-fold dispersion, so no uncertainty band is available and none is drawn. The pre-stimulus baseline (before critical-word onset at 0 ms) is shaded grey and serves as a negative control: above-chance decoding there indicates a confound and not a violation response. The cluster-level decisions shown are those of the confirmatory family, at the permutation count each artefact records.

throughout the trial, and the post-stimulus peak set against the pre-stimulus mean bounds how much separability the violation response could add. A decoder trained at one latency also classified neighbouring latencies. No null distribution was computed for that temporal-generalisation pattern, so we describe it without interpreting it. Gender agreement, whose amplitude effect was credible in the region-averaged models, produced six clusters in its pooled time-course surviving the confirmatory correction (Figure 8), spanning 378 to 418 ms, 438 to 478 ms, 530 to 602 ms, 710 to 754 ms, 850 to 926 ms, 950 to 1046 ms. Its smallest cluster-level  $p = .001$  in that pooled time-course, which is the smallest value the permutation null can return, reached by one of its clusters, and it peaked post-stimulus at AUC 0.53 (at 966 ms). Its pre-stimulus baseline sat at chance, so, unlike differential object marking, it carries no separability that precedes the critical word, and its post-stimulus clusters cannot be attributed to a difference in the region before it. Verb–object number agreement produced no clusters surviving correction in its pooled time-course (smallest cluster-level  $p = .079$  across all its analyses, before or after correction; peak overall AUC 0.53). Cross-language generalisation, in which the decoder is trained on one mini-language and tested on the other, peaked post-stimulus at AUC 0.53 for gender agreement, 0.58 for differential object marking and 0.53 for verb–object number agreement. Those values, and the temporal-generalisation ones, are descriptive. No permutation null was computed for them, so we treat them as converging context and do not read the transfer question off them. Cross-property generalisation was near chance in every direction (peak post-stimulus AUC 0.53 across all six train–test pairs). This too is descriptive. It has no permutation null, and because every participant contributes trials to all three properties, the train and test sets are not participant-independent. Its theoretical reading is taken up in the Discussion. Because each per-cell decoding score is a mean cross-validated AUC estimated on cells of differing and sometimes small participant numbers, the ordering across properties is provisional, and every cluster-level decision reported here rests on the well-populated pooled time-course (Varoquaux, 2018).

## Discussion

We set out to decide whether prior-language influence on early L3 morphosyntax is wholesale or property-by-property. The answer these data return is selective. Gender agreement, the one property with a first-language analogue, was the one whose behavioural advantage tracked that analogue, and that advantage grew credibly across sessions. The two properties novel to both prior languages showed no such advantage, and their electrophysiological modulations ran in opposite directions, which neither account predicts. Both families expect any first-language-congruent advantage to fall on gender agreement, and it did. Their predictions differ on when it should appear, which the sections below take up.

The three structures were thus acquired on three different trajectories (the full estimates are in the Results and Figure 5). Uneven trajectories alone do not decide between the accounts, since wholesale transfer constrains the initial state and says nothing about the rate at which individual properties are subsequently learned. The adjudication below rests on where and when the language effects appeared. Divergent trajectories within the same learners are in any case an established pattern. Number and gender agreement dissociate across the proficiency spectrum in second-language learners, the feature instantiated in the first language showing a violation response at lower proficiency than the novel one (Alemán Bañón et al., 2018). Brief training can also produce a developing violation response for one rule and none for another taught in the same protocol (Davidson & Indefrey, 2009). Longitudinal designs show that a P600 absent at the start of instruction can be present after a short intensive course, with its magnitude tracking behavioural sensitivity (White et al., 2012). That supports reading the present session effects as training-driven. Two features of the design nonetheless limit the reading. Session and training are confounded, so nothing in the design separates them, and every sentence frame recurs across sessions, so part of the growth of the grammaticality effect could reflect memory for particular sentences. The by-item session slopes absorb item-level change in amplitude but do not separate rule learning from sentence familiarity.

One feature of the grammaticality effect must be stated before its development is interpreted:

its polarity. With grammatical coded +0.5, a positive coefficient means that ungrammatical critical words were the more negative-going. For gender agreement that holds in every window, including 400–900 ms. That is a sustained violation negativity where our informative prior anticipated a late positivity. The data overrode that prior, and the shift under the weak priors was negligible (see *Priors*). For the two control properties, the effect is positive in the earlier windows and turns slightly negative by 400–900 ms, though that reversal is not credible in any cell. A violation negativity in learners at low proficiency has a natural reading in the developmental ERP literature, where the early stages of morphosyntactic learning are marked by N400-family negativities that give way to a P600 as proficiency grows (Osterhout et al., 2006). That literature also finds lexical ERP change within about fourteen hours of instruction in novices (McLaughlin et al., 2004), a morphosyntactic violation response within a week of brief training (Davidson & Indefrey, 2009), and individuals varying along a negativity-to-positivity continuum in native processing (Tanner & Van Hell, 2014) and in learners (Tanner et al., 2013). On that reading, the growth of the effect across sessions tracks a cohort still on the negativity side of that trajectory, consistent with the diffuse, non-posterior topography. Averaging across individuals who differ systematically in component dominance (Tanner & Van Hell, 2014) would itself flatten and diffuse the grand-average topography. The non-canonical distribution is therefore also consistent with a mixture of negativity- and positivity-dominant responders, and need not imply a single non-canonical generator.

### **Relation to the initial-exposure antecedent**

Against González Alonso et al. (2020), the closest antecedent, these data replicate the general profile but not its specifics. Two features carry over. Their learners produced no P600 to gender-agreement violations at first exposure, and neither did ours at any session. The dominant signature for gender agreement here was a violation negativity in every window, under a comparable artificial-language paradigm. The measurement occasion was not comparable, since their ERPs come from a single first-exposure session, whereas ours begin only once a learner has passed the 80% post-training criterion and continue across four sessions. An early negativity to gender-agreement violations was likewise present in both studies. The specifics do not carry over.

Their negativity was confined to the English-lexicon group and to right-anterior electrodes, whereas ours was credible on the midline, diffusely distributed, and not credibly modulated by language in any window. Their Spanish-lexicon group showed a broad positivity to violations at 300–600 ms, whereas in our gender-agreement data the ungrammatical trials were the more negative-going in that window and in the other two.

The divergences admit two kinds of explanation, and the literature on replication cautions against reaching for the substantive kind first. Substantively, the designs differ in the first language (Spanish against Norwegian), in the realisation of gender (concord across the noun phrase against a suffixed determiner with predicate-adjective agreement), and in the measurement occasion (a single first-exposure session against a trajectory whose first ERP session follows a training gate). Statistically, their groups comprised 26 and 24 learners. Effects that reach significance in samples of that size systematically overestimate their true magnitude and can even carry the wrong sign (Gelman & Carlin, 2014). The significance filter also makes published effects look more replicable than they are (Vasishth et al., 2018), and trial-level measurement error can inflate as well as attenuate an observed effect, producing spuriously large estimates in small samples (Loken & Gelman, 2017). ERP effects compound the problem, since even systematic resampling finds the magnitude, onset and scalp distribution of the P600 to vary widely across samples (Kim et al., 2024), so the exact topography and timing of a violation response are the quantities least likely to reproduce in any replication, ours included. On that reading, the level at which replication should be assessed is the qualitative one. There the two studies agree: early-stage learners of a miniature language respond to morphosyntactic violations with attention-sensitive negativities, and the canonical P600 is absent until proficiency has grown. The same arguments cut against over-reading our own group-specific effects, several of which rest on comparably small cells, and the Strength of evidence section holds them to that standard.

### **A signature more property-by-property than wholesale**

The two families were set out in the Introduction. The three properties behaved differently from one another: strong, broadly distributed learning for gender agreement; for differential object

marking an early grammaticality effect, possibly carrying a stimulus-driven component, that did not consolidate; and a late, language-conditioned signal for verb–object number agreement. That heterogeneity sits more naturally with the property-by-property family, though it is not decisive on its own, for the reason just given. The discriminating observations follow.

Transfer itself is distributed no more evenly than learning is. A credible source-language modulation appeared for all three properties, but in different measures and with different signs. For gender agreement, it was behavioural, advantaging the Mini-Norwegian group ( $b = 0.16$ , 95% CrI [0.09, 0.24],  $p_d > .999$ ), while its ERP counterpart stayed near zero. For the other two, it was the reverse, credible in the ERP signal and not behaviourally, and the two point in opposite directions: positive for verb–object number agreement ( $b = 0.012$ , 95% CrI [0.002, 0.023],  $p_d = .99$ ) and negative for differential object marking ( $b = -0.011$ , 95% CrI [-0.020, -0.003],  $p_d = .997$ ). This is not the pattern we predicted. The design treats both properties as controls, and neither family predicts a source-language modulation on them (see *The present study*), yet both showed one, in opposite directions. The prediction failed, and we have no account of the pair. Another ERP test of these models with a miniature language met a comparable outcome ([Pereira Soares et al., 2022](#)). The L2 Status Factor derivation in the Introduction would favour the Mini-English group on every property, which fits the differential-object-marking modulation and contradicts the other two language effects, so it does not account for the pair either. A general advantage of one training group is likewise hard to reconcile with modulations of opposite sign on the two controls. The remaining candidates, a difference in difficulty or in the amount of exposure each property had received by Session 6 and a false positive among eighteen cells, are not distinguishable in these data.

The claim that survives is narrower, and it turns on timing. Under the source selection the design supports, both families confine any advantage to gender agreement. Where they differ is its onset: present from the property's first session under wholesale projection, accruing with exposure under property-by-property transfer. The behavioural advantage carries both signatures: a credible advantage at the average session ( $b = 0.38$ , 95% CrI [0.10, 0.67],  $p_d = .99$ ) and credible growth across sessions (the three-way interaction above). The quantity the timing question needs is the

advantage evaluated at the property's first session, derived from the same draws:  $b = 0.18$ , 95% CrI  $[-0.12, 0.49]$ ,  $p_d = .89$ . That interval includes zero, so the onset test is indeterminate on its own terms. It neither establishes that the advantage was already in place when the property was first probed, which is the wholesale signature, nor rules it out, and its central tendency leans towards presence. As set out in the Introduction, growth is what the property-by-property account predicts and what wholesale projection permits, so the timing evidence favours the former without settling the contrast. As noted above, heterogeneity on its own is the weaker evidence, since a sufficiently unconstrained selective account accommodates almost any pattern (Schwartz & Sprouse, 2021).

The language factor sharpens the point. Grammatical gender, expressed in Mini-Norwegian through a suffixed determiner, recapitulates a structure of the learners' first language (Norwegian has both grammatical gender and a suffixal definite article). The Mini-English realisation, an English-like free determiner for a feature English does not encode, has no such first-language support. Accordingly, the behavioural gender-agreement effect showed a credible language modulation (the grammaticality  $\times$  session  $\times$  language interaction reported in the Results), advantaging the Mini-Norwegian group. Differential object marking and verb-object number agreement are novel to both Norwegian and English, and neither showed that first-language-aligned advantage in the behavioural development. Their ERP modulations therefore cannot be read as first-language congruence effects. The differential-object-marking one runs the other way: its grammaticality effect held up across the retention interval for the Mini-English group and not for the Mini-Norwegian one (Table B1). Both families predict that the one property with a first-language analogue is the one whose advantage follows it. What neither predicts is that the two properties without such an analogue should diverge from each other, and a single grammar projected wholesale offers no more purchase on that divergence than a property-by-property account does.

The behavioural gender advantage also bears on the L2 Status Factor specifically. On the derivation set out in the Introduction, an L2 default predicts either no group difference or an advantage for Mini-English, never one for Mini-Norwegian. The credible Mini-Norwegian advantage on gender agreement therefore runs directly against an L2 default source in this

population, a stronger disconfirmation than a violated null prediction.

Within the property-by-property family, the data also begin to discriminate. The Linguistic Proximity Model and the Scalpel Model both allow influence that is property-specific and, where the match is misleading, non-facilitative, and are compatible with everything observed. The Cumulative-Enhancement Model's distinctive claim, that prior languages facilitate or remain neutral, is harder to reconcile with a modulation whose direction opposes first-language congruence, although on properties without a source the model makes no clear commitment. We record this as a point on which the family's members differ, without treating it as a resolved contrast.

### **Change across the retention interval**

What happened to these effects over the months without exposure depends on how the retention step is modelled. Under the linear specification, the gender-agreement grammaticality effect was larger at Session 6 than at Session 4 in both languages and to almost the same degree (midline 400–900 ms, overall  $\Delta = 0.019$ , 95% CrI [0.010, 0.028],  $p_d > .999$ ; language difference Mini-Norwegian minus Mini-English  $\Delta = -0.0002$ , 95% CrI [-0.0184, 0.0179],  $p_d = .51$ ). That reading is an extrapolation, since the contrast is the fitted linear trajectory evaluated over the retention step (see Results) and so reports continued growth at the training rate. The linear specification cannot express a change specific to the interval.

The refit that frees the Session-6 offset revises both halves of it. A refit in which the retention step is free to differ from the training steps separates the two readings. For this cell the freed Session-6 offset, the deviation at retention, is  $\Delta = -0.012$ , 95% CrI [-0.048, 0.024],  $p_d = .74$ . Averaged over languages, the freed offset is credible in none of the six gender-agreement cells, so the apparent strengthening was the training trend carried across the gap. What the freed model does show, in five of the six cells, is a credible difference between the languages in how the effect changed over the interval (midline 400–900 ms, Mini-Norwegian minus Mini-English  $\Delta = 0.09$ , 95% CrI [0.01, 0.16],  $p_d = .99$ ). The Mini-Norwegian estimate is positive in every one of the six cells and the Mini-English estimate negative in every one of the six, although each group's own change is individually credible in fewer of them (Mini-Norwegian two of the six,

Mini-English one of the six). The average null therefore reflects two effects cancelling, and the direction of the divergence matters. The effect was maintained where the learners' first language supplies the same structure and eroded where it does not. On this evidence, the retention behaviour of gender agreement tracks the source grammar. We hold the conclusion at the level the estimates support, which is the group difference and not either group's separate trajectory, a distinction that comparisons of separately tested effects routinely blur (Gelman & Stern, 2006; S. Nieuwenhuis et al., 2011). Verb–object number agreement, by contrast, gained over the interval only within the Mini-Norwegian group (Table B1). The latest-introduced property, novel to both prior languages, strengthened for one training group only. That is a group difference on a property where property-level congruence predicts none, so we group it with the unexplained control modulations discussed above, separately from the gender-agreement transfer effect. The candidate explanations listed there apply here too, and we do not choose between them. Differential object marking, whose behavioural accuracy declined across the interval in both groups (Table B1), diverges by language in the freed model as gender agreement does, but in the opposite direction. Its retention estimate is negative within Mini-Norwegian in every one of the six cells, and the language difference is credible in five of the six cells (midline 400–900 ms, Mini-Norwegian minus Mini-English  $\Delta = -0.16$ , 95% CrI  $[-0.25, -0.06]$ ,  $p_d = .999$ ). That is the same reversal of first-language congruence that its late-window transfer term shows, which we could not account for there and cannot account for here. This property also has a stimulus confound the others do not (see Results). We therefore do not read its retention profile as consolidation of a durable representation.

Consolidation is defined as the continued transformation of an experience-dependent representation after acquisition, through recurrent reactivation during wake and sleep and redistribution of the representation to additional sites (Dudai et al., 2015), so change across an interval containing no further training cannot be a practice effect. The complementary-systems account supplies the mechanism for language specifically: rapid hippocampally supported familiarisation followed by slow neocortical consolidation, with cortical responses becoming knowledge-like only where a consolidation opportunity intervenes (M. H. Davis & Gaskell, 2009).

In the closest precedent, adults trained on an artificial language showed ERP responses that grew more native-like after months without exposure while behaviour was unchanged ([Morgan-Short, Finger, et al., 2012](#)), and responses to newly learned words shift measurably across a single overnight interval ([Bakker et al., 2015](#)). The stabilisation-versus-enhancement distinction drawn in the procedural literature ([Robertson et al., 2004](#)) frames the profiles here, once the freed estimates replace the extrapolated ones. Gender agreement shows neither enhancement nor decay on average but a divergence between the languages, verb–object number agreement gained in Mini-Norwegian alone, and differential object marking diverged in the opposite direction to gender agreement. An offline gain of this kind plausibly reflects rule abstraction ([I. L. C. Nieuwenhuis et al., 2013](#); [Tamminen et al., 2015](#)), though Mirković and Gaskell (2016) found no sleep benefit for the systematic aspects of a comparable artificial gender language. That study used a nap-length interval and behavioural measures, which a longer, electrophysiologically measured interval may reach past without contradicting.

### **Convergence and divergence of the two measures**

The electrophysiological and behavioural measures converge for gender agreement in indexing learning, and, on the derivation that bears on consolidation, at retention as well. The behavioural gain was carried by the Mini-Norwegian group, and the freed electrophysiological estimate shows the same language-dependent divergence, with no credible average change but a credible difference between the languages in the same direction. It was the linear extrapolation, which reads the retention step off the training trend, that made the electrophysiological pattern look language-general.

The two measures diverge more sharply for differential object marking. It elicited a large, credible behavioural grammaticality effect (Results) even as its ERP grammaticality effect was credible in a single early cell, and that behavioural effect then eroded at retention, a dissociation suggesting explicit, strategy-based judgement of a structure that never became an automatic processing routine. The two are not redundant, even though they are recorded on the same trials. Where they part company, they separate the automatic neural response to a violation from the

explicit accuracy of the judgement made about it.

For gender agreement, the dissociation runs the other way. The source-language advantage was carried by the judgement and not by the electrophysiological response, which suggests that at this stage first-language congruence shapes the explicit decision more than the automatic violation response. Where in processing transfer takes effect is a question the property-by-property models do not yet specify, and this pattern indicates it is separable from whether transfer occurs. An EEG study of natural-language L3 learners reports a different division, with first-language similarity consolidating implicit knowledge and second-language similarity enhancing metalinguistic judgement (Xu et al., 2024). Here the first-language-congruent advantage surfaced in the judgement, so the locus may vary with the stage of learning as well as with the source.

One behavioural result stands apart from the dissociations. Overall accuracy fell across sessions for every property, against the improvement under mere exposure that motivated the prior on the session term (Mueller et al., 2009). The design gives two reasons. Each session adds a property, so later judgements are made against a larger grammar, and Session 6 follows months without exposure. That the grammaticality effect held or grew while the level fell indicates that the decline reflects task difficulty or a drift in response criterion, with the knowledge the effect indexes intact, although the design cannot separate these.

### **What the multivariate analysis adds**

With the confirmatory decoding settled, the univariate and multivariate analyses order the three properties in broadly the same way. Gender agreement carries an amplitude effect credible in 2 of 6 cells and reliable pattern separability. Verb–object number agreement carries neither, though its grammaticality effect changes across the retention interval. Differential object marking carries separability that its pre-stimulus baseline forbids us to read as a violation response. That agreement was not guaranteed, because the two analyses ask different questions. Window-averaged amplitude asks whether a condition shifts the mean at a set of electrodes, whereas decoding asks whether the joint sensor pattern separates single trials (Hebart & Baker, 2018; Wang & Kuperberg, 2024). A divergence in either direction would not have told us which property is better represented, and

classification accuracy is in any case not a standardised effect size and cannot be compared across properties as though it were (Hebart & Baker, 2018). One limit is worth setting out before the mechanisms. Rejecting a pooled cluster null licenses only that a grammaticality pattern is decodable in some part of the sample, leaving its prevalence across participants unestimated (Allefeld et al., 2016).

One profile in these data has a documented mechanism, and the profile that would have raised a puzzle does not arise. A contrast can be decodable with little univariate effect because decoding tolerates non-uniform response signs, so neighbouring sensors of opposite polarity both carry information where averaging cancels them (Hebart & Baker, 2018). Forward modelling shows that such cancellation is substantial at the scalp for distributed sources with partly opposing orientations (Ahlfors et al., 2010). The transient mean amplitude and comparatively high classification of differential object marking are therefore compatible with a single generator whose spatial configuration is stable even where its net amplitude is small, so far as the baseline difference allows that reading. Multivariate analyses of sentence processing have repeatedly recovered effects that the corresponding ERPs did not (Cho & Brennan, 2025; He et al., 2024). The converse profile is equally well characterised. Multivariate tests draw their sensitivity from electrode-level variability in the effect within participants and are insensitive to the between-participant variance in mean amplitude that governs univariate group tests (T. Davis et al., 2014, on fMRI voxels), and increased trial-to-trial variability leaves expected mean amplitude unchanged while progressively shrinking expected decoding accuracy (Carrasco et al., 2024). An effect that is consistent in sign but spatially homogeneous would therefore yield a credible amplitude effect with comparatively little for a classifier to exploit. That profile does not describe gender agreement here, which separates reliably as well as shifting the mean. Decoding is usually the more sensitive test of the two, but it is not uniformly so, and comparable orderings are documented for the mismatch negativity and the N2pc (Carrasco et al., 2024).

These considerations constrain the claims available to us. A failure to decode is a failure to find linearly separable information under one analysis, and it leaves open whether the brain

distinguishes the conditions (Peelen & Downing, 2023). Classification performance, high or low, is a poor guide to representational content (Ritchie et al., 2019). Nor would a divergence between the two analyses, had one arisen, support a conclusion about the dimensionality of the underlying code (T. Davis et al., 2014).

The null cross-property generalisation bears on a live theoretical question. If the grammaticality response were largely a domain-general salience or integration signal (Aurnhammer et al., 2023; Sassenhagen & Fiebach, 2019), a decoder trained on one property should transfer to the others, but it transferred in no direction. That pattern is consistent with evidence that the late positivity fractionates topographically and functionally (Kaan & Swaab, 2003), that agreement features follow separable developmental courses (Alemán Bañón et al., 2018), that different violation types within the same learners are predicted by different individual-difference variables (E. S. Nichols & Joanisse, 2019), and that decoders of grammatical features generalise selectively (Cho & Brennan, 2025). Reliable cross-context transfer of linear EEG decoders is achievable (Zhao et al., 2025), which weighs against reading the present null as simple insensitivity. Even so, failed cross-classification is confounded with feature selection, since the sensors and latencies that best separate grammatical from ungrammatical trials within a property need not be those carrying a property-general signal (Kaplan et al., 2015). We therefore read the result as consistent with construction-specific coding, without taking it to refute any shared component. A Bayes-factor treatment of the decoding time-course would allow such nulls to be quantified as well as reported (Keysers et al., 2020; Teichmann, 2022).

### **Strength of evidence**

The conclusions below should be read on the graded scale set out under *Inference* (Kruschke & Liddell, 2018; Makowski et al., 2019). The gender-agreement learning effect is credible in every cell but one under both random-effect structures. The between-language difference in how that effect changed across the retention interval is credible in five of the six cells, with lower interval bounds close to zero in some of them (Table B2), so it is credible on the stated criterion without being a large effect. The average departure from the training trend at Session 6 is not. The learning

effect in particular is the most structure-independent result in the paper. The grammaticality  $\times$  session interaction is credible in 5 of 6 cells under the maximal random-effect structure and in the same 5 of 6 without the by-item slope. The verb–object number development and its Norwegian-specific retention are better described as suggestive and warrant confirmation, and the same holds for the differential-object-marking transfer interaction. It is credible in 2 of 6 cells, unchanged when the model is refitted under weakly informative priors, and it survived the move to the maximal structure. It is nonetheless small, and it sits close enough to the threshold that data-handling decisions taken before any model was fitted could bear on it. One such decision applies here: four Session-3 datasets were mis-filtered at acquisition and are excluded from every fit reported, an exclusion justified a priori and applied to the whole grid. Because no full-sample refit was run, the effect of retaining those datasets is not estimated.

The decoding evidence grades separately, and unevenly across the three properties. Gender agreement’s six clusters surviving the confirmatory correction rest on the full permutation null, on a pre-stimulus baseline at chance, and are secure on that basis. Differential object marking’s two clusters are not comparable evidence: both begin before the critical word and its highest AUC falls in the baseline, so they bound the separability a violation response could contribute without establishing that it contributes any. The session-resolved breakdowns enter no correction family and are graded no higher than descriptive. The near-absence of a gender-agreement transfer interaction in the ERP signal is a well-estimated value near zero, though it remains conditional on the prior and the interval width. The verb–object number and differential-object-marking signals therefore remain provisional, and the gender-agreement pattern is the secure result. The property-by-property reading is the account the heterogeneity supports, short of a decisive refutation of wholesale transfer, which a single study cannot deliver. The transfer interactions that remain near zero are best understood as absence of evidence. They do not demonstrate absence. We did not adopt an equivalence (region-of-practical-equivalence) criterion, so a precisely estimated coefficient whose direction is undetermined ( $p_d \approx .50$ ) marks a well-localised null estimate ([Amrhein et al., 2019](#); [Kruschke & Liddell, 2018](#)).

### Limitations and future directions

The strongest constraint on what this design can conclude is how narrowly the two families of account are separated by it. Only gender agreement has a first-language analogue, so only gender agreement places the two miniature languages in different relations to a prior grammar. The other two properties are novel to Norwegian and English alike and serve as controls. The wholesale-versus-property-by-property contrast therefore rests on a single dissociating property, which is thin ground for a claim about the unit of transfer, however the other evidence falls. A design contrasting two candidate source grammars, or several properties each aligned with a different prior language, would bear the weight this one is asked to. The heterogeneity we do observe, which runs across properties and measures and includes a reversal of sign, is more readily accommodated by a property-by-property account than by a wholesale one, but consistency with that family is not a decisive test between them. The two electrophysiological modulations on properties with no first-language analogue are predicted by neither.

The artificial-language design also buys experimental control at some cost to ecological validity, and three further features of it constrain the present conclusions. First, the properties enter at different sessions (gender agreement from Session 2, differential object marking from Session 3, verb–object number agreement from Session 4), so a property’s identity is confounded with its amount of exposure. Cross-property comparisons of learning and transfer are therefore not exposure-matched, and the weaker signatures of the later-introduced properties are in part a difference in training time. An exposure-matched design that introduced the properties in counterbalanced order would separate the two. Second, the additional gendered or case-marking languages noted under *Participants* were not screened out. Such experience could support the gender or marking structures, and it is only partially absorbed by the L1-dominance covariate, which indexes the standing of the first language without capturing the breadth of the repertoire, and which moreover enters the ERP models only. The behavioural models have no such control, so the behavioural gender advantage is the estimate most exposed to it. Third, effects are quantified as mean amplitudes in fixed latency windows, with no peak or topographic decomposition. The

absence of a native-speaker ERP baseline also means that the results speak to the development of the grammaticality response and leave its native-likeness untested. A native-speaker baseline would test whether the developing response converges on the native pattern, and a longer interval whether the divergence between the languages at retention widens. How far the priors and the interval widths condition the transfer interactions that remain near zero is addressed under *Strength of evidence*.

Two features of the sample bear on the retention inference, and a third on every between-group comparison. Of the 65 learners enrolled, 36 attended the Session 6 retention test, so the Session 6 estimates draw on the returners alone while the training trend draws on every session, and the models treat the missing sessions as ignorable given each participant's random effects. Returners may differ from those who left, and a retention estimate conditioned on returning need not describe the cohort that began. The 80% training gate described under *Transparency, preregistration and analysis status* adds a second selection, since it conditions which participant-sessions contribute data on performance in that session, and its outcomes are folded into the attendance counts. Assignment to language followed a fixed rule on the participant identifier, so the groups were not formed by random allocation, and the between-group comparisons carry the uncertainty that attends a non-randomised design.

The behavioural judgement task also lacks a stable cross-participant item factor, so item variance is absorbed into the residual for accuracy (though not for the ERP models, which use fixed sentence frames). Cognitive individual-difference predictors (working memory, inhibitory control, statistical-learning ability) are excluded here by design and analysed in a companion paper ([Bernabeu et al., 2026](#)), keeping the present focus on the linguistic transfer signal. A natural further step for the decoding strand is representational similarity analysis ([Kriegeskorte et al., 2008](#)), which would characterise what the distributed pattern represents: whether the same neural geometry that separates grammatical from ungrammatical trials in one property recurs in another. That would be a stronger test of shared representation than the cross-property generalisation reported here.

## Conclusion

Tracked from the first post-training session to a post-consolidation retention test, three morphosyntactic properties followed three different trajectories, and the one advantage that tracked a first language was confined to the one property that had a first-language analogue. On these data, transfer is better described as property-specific than as the projection of a whole grammar. We offer that conclusion provisionally. Only gender agreement dissociates the two accounts here, and under the design's own source selection, the wholesale alternative is disfavoured more by the advantage's growth over sessions than by its location.

## Data and code availability

Data extraction, model fitting and reporting are fully scripted as a reproducible research compendium ([Marwick et al., 2018](#)). Models were fitted on High-Performance Computing (HPC) clusters at Sigma2 (Norway) and the University of Oxford's Advanced Research Computing facility, on Linux (x86\_64-pc-linux-gnu) under R 4.5.1 ([R Core Team, 2025](#)), with brms 2.23.0 ([Bürkner, 2017](#)) driving CmdStan 2.39.0 through the cmdstanr 0.9.0 backend. The clusters supply that stack through environment modules, with no container. The compiler settings used on those clusters are recorded in the repository and do not affect the posterior draws. Every version given here comes from the provenance record the fitting run wrote. Model fitting used the fixed seed 20260607, set in the shared Bayesian settings and recorded in that provenance record. The decoding permutations used the seed 20260702, exported by the job script that runs them and recorded by that script beside the decoding artefacts. This manuscript reads only the script-generated summary tables, so every statistic in the Results is populated from them and none is transcribed by hand. The pooled posterior summaries, retention contrasts, convergence diagnostics, sample-flow counts and decoding time-courses that populate the figures and tables are all published with the repository. The decoding artefacts include the participant count behind every per-cell score. Figures are drawn with ggplot2 ([Wickham, 2016](#)) using the theme and colour system of depictr ([Bernabeu, 2026a](#)), whose default qualitative palette is the colourblind-safe Okabe–Ito set. The companion paper ([Bernabeu et al., 2026](#)) uses the same system, so the two share one visual identity. The literature counts in Figure 1

were retrieved from the Scopus Search API with `scopusflow` (Bernabeu, 2026b), which records the exact query beside each count, so the search is reproducible from the repository. Its coverage limits are stated in the figure caption. All extraction, modelling and reporting scripts are archived with the data in the project's Open Science Framework repository at <https://osf.io/tq7vy>, together with the environment specification: an `renv` lockfile pinning the R packages of the fitting library, and a provenance record written at fit time. That record holds the software versions in force when the ERP models were fitted, which is what the versions above are read from. The accuracy models and the decoding analyses run in the same cluster environment under separate scripts. The accuracy script records the R, brms, cmdstanr and CmdStan versions of each fit in the pooled fit metadata that accompanies the results, and the decoding job records its seed and permutation settings beside the decoding artefacts. The raw electroencephalographic recordings, too large to distribute through an ordinary version-control repository, are held in the same Open Science Framework repository as the code. The paper's analysis code and other open materials are also available at <https://github.com/pablobernabeu/LESS-morphosyntactic-transfer>.

### **Declaration of generative AI and AI-assisted technologies in the manuscript preparation process**

During the preparation of this work, the authors used Claude Code (version 2.1, Anthropic), running primarily the Claude Opus and Claude Fable models, in order to support the research, the analyses and the writing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

### **CRedit authorship contribution statement**

**Pablo Bernabeu:** Conceptualization, Methodology, Software, Formal analysis, Data curation, Visualization, Writing – original draft, Writing – review & editing. **Christina Athanasiadi:** Data curation, Project administration, Writing – review & editing. **Stella L. Pischinger:** Data curation, Writing – review & editing. **Gabriella Silva:** Conceptualization, Methodology, Investigation, Writing – review & editing. **My Ngoc Giang Hoang:** Data curation, Writing – review & editing. **Vincent DeLuca:** Conceptualization, Methodology, Supervision,

Writing – review & editing. **Jason Rothman**: Conceptualization, Funding acquisition, Resources, Supervision, Writing – review & editing. **Iva Ivanova**: Conceptualization, Methodology, Writing – review & editing. **Claudia Poch**: Methodology, Validation, Writing – review & editing. **Jorge González Alonso**: Conceptualization, Methodology, Funding acquisition, Project administration, Supervision, Writing – review & editing.

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## Appendix A

### Analytic decisions in detail

This appendix sets out the reasoning behind two analytic choices that the Method states but does not argue in full: the retention of the by-item grammaticality slope in the ERP models, and the definition of the false-discovery-rate correction family for the decoding analyses.

#### The by-item grammaticality slope

The reason for keeping the by-item grammaticality slope is that dropping it would bias the test in the direction of the hypothesis. Barr et al. (2013) show that omitting random-effect structure that is present in the population inflates the Type I error rate. They add that, for within-items manipulations specifically, an intercept-only by-item structure can inflate it severely, whatever method is used to obtain *p*-values or intervals. Grammaticality is a within-item manipulation here, so an item term carrying only an intercept and a session slope is exactly that case for the effect under test.

Matuschek et al. (2017) argue the other side: a maximal model can cost power, and they recommend reducing to a structure the data support. We report the maximal structure nonetheless, for two reasons. Their argument is that a maximal structure the data do not support costs power, a result they establish in simulations of models that converge, so it does not turn on convergence failure. They themselves note that maximal models usually do converge when fitted in a Bayesian framework such as Stan, while explicitly leaving open whether fitting overparameterised maximal models makes sense there. Ours converged, meeting the same criteria as every other model reported here. We also fit the whole grid a second time without the by-item grammaticality slope, so a reader can see which conclusions depend on it. That comparison follows.

#### Sensitivity to the by-item grammaticality slope

To show what the maximal structure buys, we refitted the whole grid without the by-item grammaticality slope, the structure a design in which grammaticality varied only within participants would have justified. All eighteen cells met the same convergence criteria under both. The consequence differs sharply between the interaction terms and the main effect, so we separate them.

The transfer conclusions do not depend on the choice. The interaction is credible for differential object marking and verb–object number agreement under both structures, and its intervals are essentially identical in width (1.00 times the simpler structure’s width at the median). The maximal structure is in fact the more generous of the two here, since one further cell reaches credibility under it (differential object marking, 300–600 ms lateral). The count is 3 of 18 cells against 2 of 18 without the slope.

The grammaticality main effect is a different matter, and it is where the simpler structure would have misled. Posterior medians barely move between the two, but the maximal structure’s intervals are about 1.13 times as wide as the simpler structure’s, and that widening is enough to change how many cells clear the credibility threshold. For gender agreement, the count is 2 of 6 cells under the maximal structure against 5 of 6 without the slope, and, for differential object marking, 1 of 6 against 2 of 6. By-item variability in the grammaticality response is therefore not negligible in these data. Had we reported the simpler structure, we would have described the gender-agreement grammaticality effect as credible across nearly the whole grid. Under the structure the design justifies, it keeps its sign and magnitude but is credible in a minority of cells. The structure-dependent claim is therefore how widely the effect is distributed, while the existence and direction of the effect hold under both.

### **The Session 3 filter exclusion**

The Method excludes the Session 3 recordings of four participants, which were high-pass filtered at 1 Hz where every other recording was filtered at 0.1 Hz. We did not assess whether these participants’ data look unusual, and the exclusion does not rest on that. It rests on the setting itself, which is a documented generator of artefacts in this kind of paradigm. Filtering data from a syntactic- and semantic-violation experiment at cutoffs of 0.3 Hz and above introduces spurious deflections of opposite polarity immediately before the true N400 and P600 effects ([Tanner et al., 2015](#)), and 1 Hz lies well inside that range. The general principle, that a high-pass cutoff must be matched to the analysis window because an inappropriate one generates artefactual slow deflections, is set out by Widmann et al. ([2015](#)). Because the exclusion follows from the filter’s known

behaviour, and nothing observed in these participants bore on it, it is not a results-contingent choice.

### **The decoding correction family**

The Method states how the second layer of multiplicity control on the decoding analyses is defined. Correcting instead across the pooled, per-session and per-session-by-language cells of one property would span nested, overlapping re-analyses of the same trials. Across those, the number of hypotheses is not well defined, and the positive-dependence condition that licenses

Benjamini–Hochberg under dependence cannot be established. A false-discovery rate computed over such a family is not interpretable. The three properties’ pooled time-courses rest on disjoint sets of trials, and correcting each property’s family separately keeps every family self-contained.

The confirmatory decisions reported here rest on the family just defined and on the full permutation null. Only the exploratory breakdowns for gender agreement still carry the coarser null of the earlier run, and no decision in the confirmatory family depends on them.

## Appendix B

### Retention contrasts in full

The Results report the retention contrasts that carry the argument. The two tables below give the full record: every Session 4 to Session 6 contrast of the reported ERP and accuracy models (Table B1), and every retention-free contrast from the refits that carry an explicit retention indicator (Table B2). Both are read from the pooled contrast artefact and reproduce with it.

**Table B1**

*Session 4 to Session 6 Retention Contrasts*

| Property | Window     |         | Measure | Contrast          | Median  | 95% CrI           | $p_d$  |
|----------|------------|---------|---------|-------------------|---------|-------------------|--------|
|          | (ms)       | Region  |         |                   |         |                   |        |
| Gender   | 200 to 500 | Lateral | ERP     | Overall           | 0.0114  | [0.0056, 0.0172]  | > .999 |
| Gender   | 200 to 500 | Lateral | ERP     | Mini Norwegian    | 0.0105  | [0.0021, 0.0189]  | .99    |
| Gender   | 200 to 500 | Lateral | ERP     | Mini English      | 0.0123  | [0.0043, 0.0203]  | .999   |
| Gender   | 200 to 500 | Lateral | ERP     | Between languages | -0.0018 | [-0.0135, 0.0098] | .62    |
| Gender   | 200 to 500 | Midline | ERP     | Overall           | 0.0070  | [-0.0014, 0.0153] | .95    |
| Gender   | 200 to 500 | Midline | ERP     | Mini Norwegian    | 0.0135  | [0.0014, 0.0257]  | .99    |
| Gender   | 200 to 500 | Midline | ERP     | Mini English      | 0.0006  | [-0.0110, 0.0124] | .54    |
| Gender   | 200 to 500 | Midline | ERP     | Between languages | 0.0128  | [-0.0040, 0.0298] | .93    |
| Gender   | 300 to 600 | Lateral | ERP     | Overall           | 0.0125  | [0.0065, 0.0186]  | > .999 |
| Gender   | 300 to 600 | Lateral | ERP     | Mini Norwegian    | 0.0101  | [0.0013, 0.0188]  | .99    |
| Gender   | 300 to 600 | Lateral | ERP     | Mini English      | 0.0149  | [0.0065, 0.0232]  | > .999 |
| Gender   | 300 to 600 | Lateral | ERP     | Between languages | -0.0048 | [-0.0168, 0.0075] | .78    |
| Gender   | 300 to 600 | Midline | ERP     | Overall           | 0.0101  | [0.0015, 0.0189]  | .99    |
| Gender   | 300 to 600 | Midline | ERP     | Mini Norwegian    | 0.0143  | [0.0016, 0.0268]  | .99    |
| Gender   | 300 to 600 | Midline | ERP     | Mini English      | 0.0062  | [-0.0058, 0.0181] | .84    |
| Gender   | 300 to 600 | Midline | ERP     | Between languages | 0.0082  | [-0.0091, 0.0253] | .82    |
| Gender   | 400 to 900 | Lateral | ERP     | Overall           | 0.0155  | [0.0092, 0.0217]  | > .999 |
| Gender   | 400 to 900 | Lateral | ERP     | Mini Norwegian    | 0.0119  | [0.0027, 0.0208]  | .99    |
| Gender   | 400 to 900 | Lateral | ERP     | Mini English      | 0.0190  | [0.0103, 0.0277]  | > .999 |
| Gender   | 400 to 900 | Lateral | ERP     | Between languages | -0.0071 | [-0.0198, 0.0054] | .87    |

**Table B1***Session 4 to Session 6 Retention Contrasts*

| Property       | Window     |         | Measure  | Contrast          | Median  | 95% CrI            | <i>p<sub>d</sub></i> |
|----------------|------------|---------|----------|-------------------|---------|--------------------|----------------------|
|                | (ms)       | Region  |          |                   |         |                    |                      |
| Gender         | 400 to 900 | Midline | ERP      | Overall           | 0.0190  | [0.0101, 0.0280]   | > .999               |
| Gender         | 400 to 900 | Midline | ERP      | Mini Norwegian    | 0.0189  | [0.0058, 0.0321]   | .998                 |
| Gender         | 400 to 900 | Midline | ERP      | Mini English      | 0.0191  | [0.0067, 0.0316]   | .999                 |
| Gender         | 400 to 900 | Midline | ERP      | Between languages | -0.0002 | [-0.0184, 0.0179]  | .51                  |
| Gender         | n/a        | n/a     | Accuracy | Overall           | 0.19    | [0.12, 0.26]       | > .999               |
| Gender         | n/a        | n/a     | Accuracy | Mini Norwegian    | 0.34    | [0.26, 0.43]       | > .999               |
| Gender         | n/a        | n/a     | Accuracy | Mini English      | 0.04    | [-0.07, 0.14]      | .76                  |
| Gender         | n/a        | n/a     | Accuracy | Between languages | 0.30    | [0.17, 0.44]       | > .999               |
| Object marking | 200 to 500 | Lateral | ERP      | Overall           | -0.0033 | [-0.0133, 0.0067]  | .74                  |
| Object marking | 200 to 500 | Lateral | ERP      | Mini Norwegian    | -0.0130 | [-0.0274, 0.0014]  | .96                  |
| Object marking | 200 to 500 | Lateral | ERP      | Mini English      | 0.0068  | [-0.0069, 0.0204]  | .83                  |
| Object marking | 200 to 500 | Lateral | ERP      | Between languages | -0.0197 | [-0.0395, 0.0002]  | .97                  |
| Object marking | 200 to 500 | Midline | ERP      | Overall           | -0.0129 | [-0.0271, 0.0012]  | .96                  |
| Object marking | 200 to 500 | Midline | ERP      | Mini Norwegian    | -0.0177 | [-0.0385, 0.0031]  | .95                  |
| Object marking | 200 to 500 | Midline | ERP      | Mini English      | -0.0082 | [-0.0275, 0.0113]  | .80                  |
| Object marking | 200 to 500 | Midline | ERP      | Between languages | -0.0095 | [-0.0382, 0.0187]  | .74                  |
| Object marking | 300 to 600 | Lateral | ERP      | Overall           | 0.0042  | [-0.0062, 0.0147]  | .79                  |
| Object marking | 300 to 600 | Lateral | ERP      | Mini Norwegian    | -0.0072 | [-0.0222, 0.0078]  | .83                  |
| Object marking | 300 to 600 | Lateral | ERP      | Mini English      | 0.0160  | [0.0015, 0.0304]   | .99                  |
| Object marking | 300 to 600 | Lateral | ERP      | Between languages | -0.0231 | [-0.0437, -0.0026] | .99                  |
| Object marking | 300 to 600 | Midline | ERP      | Overall           | -0.0015 | [-0.0162, 0.0130]  | .58                  |
| Object marking | 300 to 600 | Midline | ERP      | Mini Norwegian    | -0.0065 | [-0.0280, 0.0146]  | .73                  |
| Object marking | 300 to 600 | Midline | ERP      | Mini English      | 0.0036  | [-0.0165, 0.0235]  | .64                  |
| Object marking | 300 to 600 | Midline | ERP      | Between languages | -0.0103 | [-0.0396, 0.0188]  | .75                  |
| Object marking | 400 to 900 | Lateral | ERP      | Overall           | 0.0064  | [-0.0042, 0.0171]  | .88                  |
| Object marking | 400 to 900 | Lateral | ERP      | Mini Norwegian    | -0.0081 | [-0.0235, 0.0073]  | .85                  |
| Object marking | 400 to 900 | Lateral | ERP      | Mini English      | 0.0214  | [0.0067, 0.0360]   | .998                 |
| Object marking | 400 to 900 | Lateral | ERP      | Between languages | -0.0295 | [-0.0507, -0.0081] | .997                 |

**Table B1***Session 4 to Session 6 Retention Contrasts*

| Property         | Window     |         | Measure  | Contrast          | Median  | 95% CrI           | <i>p<sub>d</sub></i> |
|------------------|------------|---------|----------|-------------------|---------|-------------------|----------------------|
|                  | (ms)       | Region  |          |                   |         |                   |                      |
| Object marking   | 400 to 900 | Midline | ERP      | Overall           | -0.0022 | [-0.0172, 0.0129] | .61                  |
| Object marking   | 400 to 900 | Midline | ERP      | Mini Norwegian    | -0.0105 | [-0.0319, 0.0112] | .83                  |
| Object marking   | 400 to 900 | Midline | ERP      | Mini English      | 0.0063  | [-0.0143, 0.0271] | .73                  |
| Object marking   | 400 to 900 | Midline | ERP      | Between languages | -0.0169 | [-0.0465, 0.0129] | .86                  |
| Object marking   | n/a        | n/a     | Accuracy | Overall           | -0.31   | [-0.41, -0.21]    | > .999               |
| Object marking   | n/a        | n/a     | Accuracy | Mini Norwegian    | -0.22   | [-0.35, -0.10]    | > .999               |
| Object marking   | n/a        | n/a     | Accuracy | Mini English      | -0.40   | [-0.57, -0.24]    | > .999               |
| Object marking   | n/a        | n/a     | Accuracy | Between languages | 0.18    | [-0.02, 0.39]     | .96                  |
| Number agreement | 200 to 500 | Lateral | ERP      | Overall           | 0.0108  | [-0.0096, 0.0310] | .85                  |
| Number agreement | 200 to 500 | Lateral | ERP      | Mini Norwegian    | 0.0187  | [-0.0100, 0.0477] | .90                  |
| Number agreement | 200 to 500 | Lateral | ERP      | Mini English      | 0.0022  | [-0.0267, 0.0311] | .56                  |
| Number agreement | 200 to 500 | Lateral | ERP      | Between languages | 0.0164  | [-0.0244, 0.0575] | .79                  |
| Number agreement | 200 to 500 | Midline | ERP      | Overall           | -0.0016 | [-0.0303, 0.0273] | .54                  |
| Number agreement | 200 to 500 | Midline | ERP      | Mini Norwegian    | 0.0047  | [-0.0360, 0.0451] | .59                  |
| Number agreement | 200 to 500 | Midline | ERP      | Mini English      | -0.0081 | [-0.0485, 0.0327] | .65                  |
| Number agreement | 200 to 500 | Midline | ERP      | Between languages | 0.0127  | [-0.0449, 0.0705] | .67                  |
| Number agreement | 300 to 600 | Lateral | ERP      | Overall           | 0.0136  | [-0.0075, 0.0348] | .90                  |
| Number agreement | 300 to 600 | Lateral | ERP      | Mini Norwegian    | 0.0191  | [-0.0106, 0.0494] | .89                  |
| Number agreement | 300 to 600 | Lateral | ERP      | Mini English      | 0.0078  | [-0.0217, 0.0375] | .69                  |
| Number agreement | 300 to 600 | Lateral | ERP      | Between languages | 0.0112  | [-0.0303, 0.0539] | .70                  |
| Number agreement | 300 to 600 | Midline | ERP      | Overall           | 0.0074  | [-0.0222, 0.0365] | .68                  |
| Number agreement | 300 to 600 | Midline | ERP      | Mini Norwegian    | 0.0137  | [-0.0283, 0.0558] | .74                  |
| Number agreement | 300 to 600 | Midline | ERP      | Mini English      | 0.0005  | [-0.0414, 0.0421] | .51                  |
| Number agreement | 300 to 600 | Midline | ERP      | Between languages | 0.0135  | [-0.0460, 0.0731] | .67                  |
| Number agreement | 400 to 900 | Lateral | ERP      | Overall           | 0.0305  | [0.0084, 0.0526]  | .997                 |
| Number agreement | 400 to 900 | Lateral | ERP      | Mini Norwegian    | 0.0551  | [0.0241, 0.0866]  | > .999               |
| Number agreement | 400 to 900 | Lateral | ERP      | Mini English      | 0.0043  | [-0.0271, 0.0357] | .61                  |
| Number agreement | 400 to 900 | Lateral | ERP      | Between languages | 0.0508  | [0.0066, 0.0955]  | .99                  |

**Table B1***Session 4 to Session 6 Retention Contrasts*

| Property         | Window     |         | Measure  | Contrast          | Median | 95% CrI           | $p_d$ |
|------------------|------------|---------|----------|-------------------|--------|-------------------|-------|
|                  | (ms)       | Region  |          |                   |        |                   |       |
| Number agreement | 400 to 900 | Midline | ERP      | Overall           | 0.0300 | [-0.0017, 0.0612] | .97   |
| Number agreement | 400 to 900 | Midline | ERP      | Mini Norwegian    | 0.0546 | [0.0107, 0.0988]  | .99   |
| Number agreement | 400 to 900 | Midline | ERP      | Mini English      | 0.0039 | [-0.0409, 0.0475] | .57   |
| Number agreement | 400 to 900 | Midline | ERP      | Between languages | 0.0509 | [-0.0110, 0.1138] | .95   |
| Number agreement | n/a        | n/a     | Accuracy | Overall           | -0.11  | [-0.27, 0.06]     | .89   |
| Number agreement | n/a        | n/a     | Accuracy | Mini Norwegian    | -0.02  | [-0.25, 0.21]     | .57   |
| Number agreement | n/a        | n/a     | Accuracy | Mini English      | -0.20  | [-0.44, 0.04]     | .95   |
| Number agreement | n/a        | n/a     | Accuracy | Between languages | 0.18   | [-0.15, 0.51]     | .86   |

*Note.* Each contrast is the change in the grammaticality effect across the retention interval: posterior median, 95% credible interval and probability of direction, overall, within each language and for the between-language difference (Mini-Norwegian minus Mini-English). The two model families are on different scales, as the Measure column records. ERP contrasts are in within-participant SD units of amplitude and accuracy contrasts in log-odds, so medians should be compared only within a measure. Dashes in the window and region columns mark the accuracy models, which are fitted per property and have neither. For compactness, Gender means subject–verb gender agreement, Object marking means differential object marking, and Number agreement means verb–object number agreement.

**Table B2***Retention-Free Contrasts From the Refits Carrying an Explicit Retention Indicator*

| Property | Window     |         | Contrast       | Median | 95% CrI         | $p_d$ |
|----------|------------|---------|----------------|--------|-----------------|-------|
|          | (ms)       | Region  |                |        |                 |       |
| Gender   | 200 to 500 | Lateral | Overall        | 0.003  | [-0.020, 0.027] | .61   |
| Gender   | 200 to 500 | Lateral | Mini Norwegian | 0.027  | [-0.006, 0.061] | .95   |

**Table B2***Retention-Free Contrasts From the Refits Carrying an Explicit Retention Indicator*

| Property       | Window     |         | Contrast          | Median | 95% CrI          | $p_d$  |
|----------------|------------|---------|-------------------|--------|------------------|--------|
|                | (ms)       | Region  |                   |        |                  |        |
| Gender         | 200 to 500 | Lateral | Mini English      | -0.020 | [-0.053, 0.013]  | .89    |
| Gender         | 200 to 500 | Lateral | Between languages | 0.048  | [0.001, 0.095]   | .98    |
| Gender         | 200 to 500 | Midline | Overall           | 0.011  | [-0.023, 0.045]  | .73    |
| Gender         | 200 to 500 | Midline | Mini Norwegian    | 0.051  | [0.002, 0.100]   | .98    |
| Gender         | 200 to 500 | Midline | Mini English      | -0.029 | [-0.077, 0.019]  | .88    |
| Gender         | 200 to 500 | Midline | Between languages | 0.080  | [0.011, 0.149]   | .99    |
| Gender         | 300 to 600 | Lateral | Overall           | 0.001  | [-0.023, 0.026]  | .55    |
| Gender         | 300 to 600 | Lateral | Mini Norwegian    | 0.030  | [-0.005, 0.065]  | .96    |
| Gender         | 300 to 600 | Lateral | Mini English      | -0.027 | [-0.061, 0.008]  | .94    |
| Gender         | 300 to 600 | Lateral | Between languages | 0.057  | [0.008, 0.106]   | .99    |
| Gender         | 300 to 600 | Midline | Overall           | 0.006  | [-0.028, 0.041]  | .64    |
| Gender         | 300 to 600 | Midline | Mini Norwegian    | 0.058  | [0.008, 0.109]   | .99    |
| Gender         | 300 to 600 | Midline | Mini English      | -0.044 | [-0.092, 0.005]  | .96    |
| Gender         | 300 to 600 | Midline | Between languages | 0.102  | [0.031, 0.173]   | .998   |
| Gender         | 400 to 900 | Lateral | Overall           | 0.001  | [-0.025, 0.026]  | .52    |
| Gender         | 400 to 900 | Lateral | Mini Norwegian    | 0.024  | [-0.012, 0.060]  | .91    |
| Gender         | 400 to 900 | Lateral | Mini English      | -0.022 | [-0.058, 0.013]  | .89    |
| Gender         | 400 to 900 | Lateral | Between languages | 0.047  | [-0.005, 0.097]  | .96    |
| Gender         | 400 to 900 | Midline | Overall           | -0.012 | [-0.048, 0.024]  | .74    |
| Gender         | 400 to 900 | Midline | Mini Norwegian    | 0.033  | [-0.019, 0.085]  | .89    |
| Gender         | 400 to 900 | Midline | Mini English      | -0.055 | [-0.106, -0.004] | .98    |
| Gender         | 400 to 900 | Midline | Between languages | 0.088  | [0.014, 0.161]   | .99    |
| Object marking | 200 to 500 | Lateral | Overall           | -0.020 | [-0.051, 0.010]  | .90    |
| Object marking | 200 to 500 | Lateral | Mini Norwegian    | -0.064 | [-0.107, -0.022] | .999   |
| Object marking | 200 to 500 | Lateral | Mini English      | 0.025  | [-0.019, 0.070]  | .87    |
| Object marking | 200 to 500 | Lateral | Between languages | -0.090 | [-0.151, -0.028] | .998   |
| Object marking | 200 to 500 | Midline | Overall           | -0.041 | [-0.084, 0.004]  | .96    |
| Object marking | 200 to 500 | Midline | Mini Norwegian    | -0.135 | [-0.196, -0.074] | > .999 |

**Table B2***Retention-Free Contrasts From the Refits Carrying an Explicit Retention Indicator*

| Property       | Window     |         | Contrast          | Median | 95% CrI          | $p_d$  |
|----------------|------------|---------|-------------------|--------|------------------|--------|
|                | (ms)       | Region  |                   |        |                  |        |
| Object marking | 200 to 500 | Midline | Mini English      | 0.057  | [-0.005, 0.119]  | .96    |
| Object marking | 200 to 500 | Midline | Between languages | -0.191 | [-0.279, -0.104] | > .999 |
| Object marking | 300 to 600 | Lateral | Overall           | -0.020 | [-0.052, 0.012]  | .89    |
| Object marking | 300 to 600 | Lateral | Mini Norwegian    | -0.053 | [-0.098, -0.009] | .99    |
| Object marking | 300 to 600 | Lateral | Mini English      | 0.014  | [-0.031, 0.060]  | .74    |
| Object marking | 300 to 600 | Lateral | Between languages | -0.068 | [-0.131, -0.004] | .98    |
| Object marking | 300 to 600 | Midline | Overall           | -0.042 | [-0.088, 0.004]  | .96    |
| Object marking | 300 to 600 | Midline | Mini Norwegian    | -0.132 | [-0.195, -0.068] | > .999 |
| Object marking | 300 to 600 | Midline | Mini English      | 0.050  | [-0.015, 0.115]  | .94    |
| Object marking | 300 to 600 | Midline | Between languages | -0.182 | [-0.271, -0.092] | > .999 |
| Object marking | 400 to 900 | Lateral | Overall           | 0.015  | [-0.018, 0.048]  | .81    |
| Object marking | 400 to 900 | Lateral | Mini Norwegian    | -0.007 | [-0.053, 0.039]  | .62    |
| Object marking | 400 to 900 | Lateral | Mini English      | 0.038  | [-0.010, 0.084]  | .94    |
| Object marking | 400 to 900 | Lateral | Between languages | -0.045 | [-0.109, 0.022]  | .91    |
| Object marking | 400 to 900 | Midline | Overall           | 0.002  | [-0.044, 0.049]  | .54    |
| Object marking | 400 to 900 | Midline | Mini Norwegian    | -0.075 | [-0.140, -0.010] | .99    |
| Object marking | 400 to 900 | Midline | Mini English      | 0.082  | [0.016, 0.150]   | .99    |
| Object marking | 400 to 900 | Midline | Between languages | -0.157 | [-0.251, -0.063] | .999   |

*Note.* Each contrast is the deviation of the grammaticality effect at Session 6 from the trend established across the training sessions: posterior median, 95% credible interval and probability of direction, overall, within each language and for their difference (Mini-Norwegian minus Mini-English). All estimates are in within-participant SD units of amplitude. Verb–object number agreement is omitted because its indicator is not identifiable. For compactness, Gender means subject–verb gender agreement and Object marking means differential object marking.

## Appendix C

### Prior specifications

The reasoning behind each prior is given in the Method, under *Priors*. What follows is the specification itself. Every mean and standard deviation is on the standardised scale, so a coefficient is in standard-deviation units, within participant for the ERP models. Prior means encode a direction and a regularised magnitude only, never a literature effect size, because the sources do not supply a transferable one. A directional prior is placed only where the literature supports a sign. Everything else, including the grammaticality slope in the 200–500 ms window, differential object marking at 300–600 ms, the hemisphere interactions and the transfer interaction under test, takes the model's zero-centred default.

**Table C1**

*Prior Distributions Placed on Each Parameter*

| Parameter                         | Model or window                                              | Prior        |
|-----------------------------------|--------------------------------------------------------------|--------------|
| All remaining coefficients        | all ERP models                                               | N(0, 0.5)    |
| Grammaticality                    | 300–600 ms, agreement properties                             | N(-0.1, 0.3) |
| Grammaticality                    | 400–900 ms, agreement properties                             | N(-0.2, 0.3) |
| Grammaticality                    | 400–900 ms, differential object marking                      | N(-0.2, 0.4) |
| Grammaticality x caudality        | 300–600 ms, agreement properties; 400–900 ms, all properties | N(0.1, 0.3)  |
| Intercept                         | all ERP models                                               | N(0, 0.5)    |
| Random-effect correlations        | all ERP models                                               | LKJ(2)       |
| Random-effect standard deviations | all ERP models                                               | t(3, 0, 2.5) |
| Residual standard deviation       | all ERP models                                               | t(3, 0, 2.5) |
| All remaining coefficients        | accuracy models                                              | N(0, 1)      |
| Intercept                         | accuracy models                                              | N(1.5, 1)    |
| Random-effect correlations        | accuracy models                                              | LKJ(2)       |
| Random-effect standard deviations | accuracy models                                              | t(3, 0, 2.5) |
| Session                           | accuracy models                                              | N(0.3, 0.5)  |

*Note.* Zero-centred entries carry no directional claim. The standard deviation is what bounds the prior's influence. The weakly informative baseline used for the sensitivity analysis replaces every informative mean below with zero and widens the corresponding standard deviation.

## Appendix D

### Convergence diagnostics

This appendix reports the realised sampling diagnostics for every fitted model. Each of the 18 ERP cells was fitted under two random-effect structures (base and maximal) and two prior sets, the literature-derived informative priors reported throughout and the weakly informative baseline used for the sensitivity analysis, giving 72 grid fits. Each of the 3 accuracy models was fitted under the two prior sets alone, having a single random-effect structure because no by-item term is carried, giving 6 fits. The retention-free refits add 24 fits, the 12 gender-agreement and differential-object-marking cells under both structures with the informative priors. In all, 102 models were fitted. Table D1 gives the realised diagnostics for every one, so a model that converged comfortably can be told from one that converged marginally.

**Table D1**

*Convergence Diagnostics for Every Fitted Model*

| Property       | Window  | Region  | Structure | Priors      | Retention-free | $\hat{R}$ | ESS    | ESS    |
|----------------|---------|---------|-----------|-------------|----------------|-----------|--------|--------|
|                | (ms)    |         |           |             |                |           | (bulk) | (tail) |
| Object marking |         |         | base      | informative | no             | 1.001     | 5020   | 8149   |
| Object marking |         |         | base      | weak        | no             | 1.002     | 4825   | 7707   |
| Gender         |         |         | base      | informative | no             | 1.001     | 4086   | 7874   |
| Gender         |         |         | base      | weak        | no             | 1.001     | 3675   | 7274   |
| Number agr.    |         |         | base      | informative | no             | 1.001     | 5673   | 10553  |
| Number agr.    |         |         | base      | weak        | no             | 1.001     | 5156   | 9021   |
| Object marking | 200–500 | Lateral | base      | informative | no             | 1.006     | 830    | 1808   |
| Object marking | 200–500 | Lateral | maximal   | informative | no             | 1.003     | 978    | 1420   |
| Object marking | 200–500 | Lateral | maximal   | informative | yes            | 1.006     | 1105   | 1474   |
| Object marking | 200–500 | Lateral | base      | informative | yes            | 1.007     | 840    | 1519   |
| Object marking | 200–500 | Lateral | base      | weak        | no             | 1.004     | 978    | 1962   |
| Object marking | 200–500 | Lateral | maximal   | weak        | no             | 1.006     | 1038   | 1128   |
| Object marking | 200–500 | Midline | base      | informative | no             | 1.006     | 874    | 2408   |
| Object marking | 200–500 | Midline | maximal   | informative | no             | 1.005     | 865    | 2230   |

**Table D1***Convergence Diagnostics for Every Fitted Model*

| Property       | Window<br>(ms) | Region  | Structure | Priors      | Retention-free | $\hat{R}$ | ESS<br>(bulk) | ESS<br>(tail) |
|----------------|----------------|---------|-----------|-------------|----------------|-----------|---------------|---------------|
| Object marking | 200–500        | Midline | maximal   | informative | yes            | 1.007     | 1016          | 2520          |
| Object marking | 200–500        | Midline | base      | informative | yes            | 1.003     | 776           | 2204          |
| Object marking | 200–500        | Midline | base      | weak        | no             | 1.003     | 849           | 2195          |
| Object marking | 200–500        | Midline | maximal   | weak        | no             | 1.005     | 868           | 2310          |
| Object marking | 300–600        | Lateral | base      | informative | no             | 1.005     | 837           | 1835          |
| Object marking | 300–600        | Lateral | maximal   | informative | no             | 1.006     | 809           | 1671          |
| Object marking | 300–600        | Lateral | maximal   | informative | yes            | 1.003     | 732           | 1401          |
| Object marking | 300–600        | Lateral | base      | informative | yes            | 1.005     | 885           | 1909          |
| Object marking | 300–600        | Lateral | base      | weak        | no             | 1.002     | 904           | 1756          |
| Object marking | 300–600        | Lateral | maximal   | weak        | no             | 1.007     | 593           | 1200          |
| Object marking | 300–600        | Midline | base      | informative | no             | 1.003     | 1019          | 2863          |
| Object marking | 300–600        | Midline | maximal   | informative | no             | 1.005     | 1335          | 2937          |
| Object marking | 300–600        | Midline | maximal   | informative | yes            | 1.004     | 821           | 2597          |
| Object marking | 300–600        | Midline | base      | informative | yes            | 1.003     | 1134          | 3108          |
| Object marking | 300–600        | Midline | base      | weak        | no             | 1.004     | 875           | 2837          |
| Object marking | 300–600        | Midline | maximal   | weak        | no             | 1.004     | 1135          | 3310          |
| Object marking | 400–900        | Lateral | base      | informative | no             | 1.001     | 4547          | 7922          |
| Object marking | 400–900        | Lateral | maximal   | informative | no             | 1.002     | 3782          | 6859          |
| Object marking | 400–900        | Lateral | maximal   | informative | yes            | 1.001     | 5121          | 8807          |
| Object marking | 400–900        | Lateral | base      | informative | yes            | 1.001     | 5186          | 10706         |
| Object marking | 400–900        | Lateral | base      | weak        | no             | 1.001     | 4777          | 8250          |
| Object marking | 400–900        | Lateral | maximal   | weak        | no             | 1.001     | 3511          | 7499          |
| Object marking | 400–900        | Midline | base      | informative | no             | 1.006     | 1239          | 1715          |
| Object marking | 400–900        | Midline | maximal   | informative | no             | 1.002     | 1253          | 1612          |
| Object marking | 400–900        | Midline | maximal   | informative | yes            | 1.003     | 1927          | 1615          |
| Object marking | 400–900        | Midline | base      | informative | yes            | 1.001     | 1835          | 2101          |
| Object marking | 400–900        | Midline | base      | weak        | no             | 1.003     | 1206          | 1241          |
| Object marking | 400–900        | Midline | maximal   | weak        | no             | 1.003     | 1247          | 1456          |

**Table D1**  
*Convergence Diagnostics for Every Fitted Model*

| Property | Window<br>(ms) | Region  | Structure | Priors      | Retention-free | $\hat{R}$ | ESS<br>(bulk) | ESS<br>(tail) |
|----------|----------------|---------|-----------|-------------|----------------|-----------|---------------|---------------|
| Gender   | 200–500        | Lateral | base      | informative | no             | 1.001     | 5641          | 11270         |
| Gender   | 200–500        | Lateral | maximal   | informative | no             | 1.001     | 5281          | 10703         |
| Gender   | 200–500        | Lateral | maximal   | informative | yes            | 1.001     | 5031          | 9582          |
| Gender   | 200–500        | Lateral | base      | informative | yes            | 1.001     | 5070          | 9953          |
| Gender   | 200–500        | Lateral | base      | weak        | no             | 1.001     | 5998          | 11427         |
| Gender   | 200–500        | Lateral | maximal   | weak        | no             | 1.001     | 5835          | 12599         |
| Gender   | 200–500        | Midline | base      | informative | no             | 1.004     | 2105          | 2281          |
| Gender   | 200–500        | Midline | maximal   | informative | no             | 1.002     | 2354          | 2691          |
| Gender   | 200–500        | Midline | maximal   | informative | yes            | 1.002     | 2286          | 2330          |
| Gender   | 200–500        | Midline | base      | informative | yes            | 1.003     | 2180          | 2741          |
| Gender   | 200–500        | Midline | base      | weak        | no             | 1.004     | 2216          | 2268          |
| Gender   | 200–500        | Midline | maximal   | weak        | no             | 1.002     | 2771          | 3133          |
| Gender   | 300–600        | Lateral | base      | informative | no             | 1.001     | 4671          | 9191          |
| Gender   | 300–600        | Lateral | maximal   | informative | no             | 1.001     | 4474          | 9294          |
| Gender   | 300–600        | Lateral | maximal   | informative | yes            | 1.001     | 3870          | 7376          |
| Gender   | 300–600        | Lateral | base      | informative | yes            | 1.001     | 4392          | 8832          |
| Gender   | 300–600        | Lateral | base      | weak        | no             | 1.001     | 5124          | 10134         |
| Gender   | 300–600        | Lateral | maximal   | weak        | no             | 1.001     | 4516          | 9209          |
| Gender   | 300–600        | Midline | base      | informative | no             | 1.002     | 2715          | 4016          |
| Gender   | 300–600        | Midline | maximal   | informative | no             | 1.003     | 2413          | 3324          |
| Gender   | 300–600        | Midline | maximal   | informative | yes            | 1.002     | 2398          | 2298          |
| Gender   | 300–600        | Midline | base      | informative | yes            | 1.002     | 2479          | 3397          |
| Gender   | 300–600        | Midline | base      | weak        | no             | 1.001     | 3082          | 3811          |
| Gender   | 300–600        | Midline | maximal   | weak        | no             | 1.001     | 2644          | 4045          |
| Gender   | 400–900        | Lateral | base      | informative | no             | 1.005     | 1640          | 1306          |
| Gender   | 400–900        | Lateral | maximal   | informative | no             | 1.002     | 2170          | 3454          |
| Gender   | 400–900        | Lateral | maximal   | informative | yes            | 1.002     | 1610          | 2320          |
| Gender   | 400–900        | Lateral | base      | informative | yes            | 1.003     | 1146          | 1149          |

**Table D1**  
*Convergence Diagnostics for Every Fitted Model*

| Property    | Window<br>(ms) | Region  | Structure | Priors      | Retention-free | $\hat{R}$ | ESS<br>(bulk) | ESS<br>(tail) |
|-------------|----------------|---------|-----------|-------------|----------------|-----------|---------------|---------------|
| Gender      | 400–900        | Lateral | base      | weak        | no             | 1.004     | 1596          | 1974          |
| Gender      | 400–900        | Lateral | maximal   | weak        | no             | 1.003     | 1868          | 2776          |
| Gender      | 400–900        | Midline | base      | informative | no             | 1.006     | 766           | 1215          |
| Gender      | 400–900        | Midline | maximal   | informative | no             | 1.003     | 957           | 1438          |
| Gender      | 400–900        | Midline | maximal   | informative | yes            | 1.007     | 813           | 1423          |
| Gender      | 400–900        | Midline | base      | informative | yes            | 1.004     | 983           | 1246          |
| Gender      | 400–900        | Midline | base      | weak        | no             | 1.007     | 1046          | 1637          |
| Gender      | 400–900        | Midline | maximal   | weak        | no             | 1.005     | 759           | 1354          |
| Number agr. | 200–500        | Lateral | base      | informative | no             | 1.003     | 1381          | 2376          |
| Number agr. | 200–500        | Lateral | maximal   | informative | no             | 1.002     | 1447          | 2336          |
| Number agr. | 200–500        | Lateral | base      | weak        | no             | 1.003     | 1729          | 2773          |
| Number agr. | 200–500        | Lateral | maximal   | weak        | no             | 1.003     | 1493          | 2217          |
| Number agr. | 200–500        | Midline | base      | informative | no             | 1.002     | 1717          | 2770          |
| Number agr. | 200–500        | Midline | maximal   | informative | no             | 1.004     | 1999          | 3190          |
| Number agr. | 200–500        | Midline | base      | weak        | no             | 1.005     | 1153          | 2235          |
| Number agr. | 200–500        | Midline | maximal   | weak        | no             | 1.002     | 2219          | 3603          |
| Number agr. | 300–600        | Lateral | base      | informative | no             | 1.004     | 1171          | 2421          |
| Number agr. | 300–600        | Lateral | maximal   | informative | no             | 1.003     | 1178          | 2354          |
| Number agr. | 300–600        | Lateral | base      | weak        | no             | 1.003     | 1061          | 2399          |
| Number agr. | 300–600        | Lateral | maximal   | weak        | no             | 1.006     | 1083          | 2564          |
| Number agr. | 300–600        | Midline | base      | informative | no             | 1.005     | 1105          | 2251          |
| Number agr. | 300–600        | Midline | maximal   | informative | no             | 1.004     | 1396          | 2487          |
| Number agr. | 300–600        | Midline | base      | weak        | no             | 1.004     | 1029          | 1824          |
| Number agr. | 300–600        | Midline | maximal   | weak        | no             | 1.007     | 1359          | 2303          |
| Number agr. | 400–900        | Lateral | base      | informative | no             | 1.002     | 1291          | 2121          |
| Number agr. | 400–900        | Lateral | maximal   | informative | no             | 1.002     | 1067          | 2151          |
| Number agr. | 400–900        | Lateral | base      | weak        | no             | 1.006     | 901           | 2209          |
| Number agr. | 400–900        | Lateral | maximal   | weak        | no             | 1.007     | 825           | 2065          |

**Table D1***Convergence Diagnostics for Every Fitted Model*

| Property    | Window  | Region  | Structure | Priors      | Retention-free | $\hat{R}$ | ESS    | ESS    |
|-------------|---------|---------|-----------|-------------|----------------|-----------|--------|--------|
|             | (ms)    |         |           |             |                |           | (bulk) | (tail) |
| Number agr. | 400–900 | Midline | base      | informative | no             | 1.003     | 1321   | 2318   |
| Number agr. | 400–900 | Midline | maximal   | informative | no             | 1.002     | 1228   | 2234   |
| Number agr. | 400–900 | Midline | base      | weak        | no             | 1.002     | 1264   | 2104   |
| Number agr. | 400–900 | Midline | maximal   | weak        | no             | 1.003     | 1138   | 2147   |

*Note.* Properties are abbreviated to keep the table within the measure: Gender is subject–verb gender agreement, Object marking is differential object marking, and Number agreement is verb–object number agreement.  $\hat{R}$  is the maximum rank-normalised  $\hat{R}$  and each ESS the minimum effective sample size, both taken over all parameters of the model. A blank window and region mark the grammaticality-judgement accuracy models, which are not resolved by time window or scalp region. The retention-free column marks the refits in which the Session-6 step is freed from the training trend. No model produced a divergent transition, and all met the criteria of  $\hat{R}$  below 1.01 with both effective sample sizes above 400.

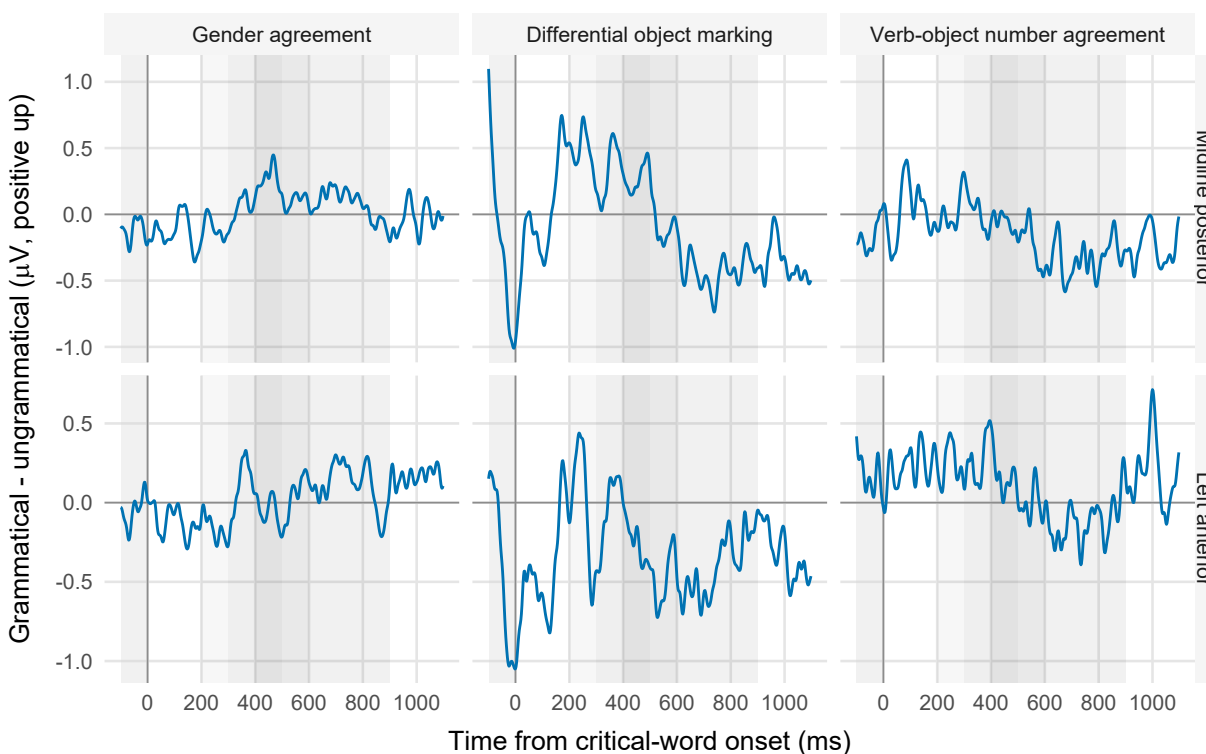
## Appendix E

### Grand-average difference waves

Figure E1 plots the grand-average difference waves (grammatical minus ungrammatical mean amplitude) at the same two sites as Figure 6. The two displays summarise the same condition averages, so the difference waves are placed in this appendix, where they aid legibility without adding information. Isolating the difference makes the grammaticality effect visible where the overlaid condition waveforms are dominated by the shared evoked response.

**Figure E1**

*Grand-Average Difference Waves per Property at the Two Representative Sites*



*Note.* Difference waves are grammatical minus ungrammatical mean amplitude at the same two sites as the grand-average waveforms. Values above zero indicate a more positive amplitude on grammatical trials. Time zero is critical-word onset and the pre-stimulus baseline (−100 to 0 ms) is shaded, with window shading as in the waveform figure. The vertical axis is scaled separately for the two sites. These are descriptive condition averages with no uncertainty band, and the inferential statements come from the models. Like the waveforms, they are trial-weighted and computed on the full sample, including the four mis-filtered Session-3 datasets.